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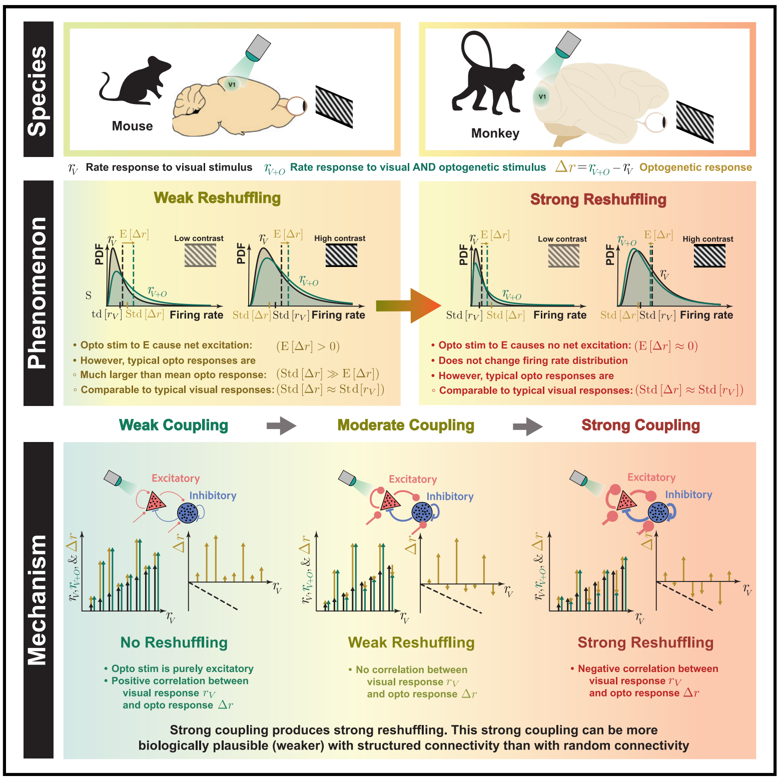
Mechanisms underlying reshuffling of visual responses by optogenetic stimulation in mice and monkeys

Graphical abstract



Highlights

- In mouse and monkey V1, optogenetic stimulation of E cells reshuffles firing rates
- Response statistics form a continuum, mice/monkeys lying in the low-/high-rate regions
- Model networks produce reshuffling if coupling is strong and optogenetic input weak
- Structured connectivity lessens required coupling strength and optogenetic weakness

Authors

Alessandro Sanzeni,
Agostina Palmigiano,
Tuan H. Nguyen, ..., Mark H. Histed,
Kenneth D. Miller, Nicolas Brunel

Correspondence

ken@neurotheory.columbia.edu (K.D.M.),
nicolas.brunel@duke.edu (N.B.)

In brief

Sanzeni et al. find that in both mouse and monkey V1, optogenetic stimulation of excitatory cells strongly changes individual neuronal firing rates but not their distribution: "rate reshuffling." Network models reproduce reshuffling given sufficiently strong interactions between neurons. The results demonstrate the importance of recurrent interactions in shaping network response.

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Mechanisms underlying reshuffling of visual responses by optogenetic stimulation in mice and monkeys

Alessandro Sanzeni,[1,2,3,9] Agostina Palmigiano,[2,9] Tuan H. Nguyen,[2,4,9] Junxiang Luo,[5] Jonathan J. Nassi,[5] John H. Reynolds,[5] Mark H. Histed,[6] Kenneth D. Miller,[2,7,10,*] and Nicolas Brunel[3,8,10,11,*]

[1]Department of Computing Sciences, Bocconi University, 20100 Milan, Italy
[2]Center for Theoretical Neuroscience and Mortimer B Zuckerman Mind Brain Behavior Institute, Columbia University, New York, NY 10027, USA
[3]Department of Neurobiology, Duke University, Durham, NC 27710, USA
[4]Department of Physics, Columbia University, New York, NY 10027, USA
[5]Systems Neurobiology Laboratory, The Salk Institute for Biological Studies, La Jolla, CA 92037, USA
[6]National Institute of Mental Health Intramural Program, NIH, Bethesda, MD 20814, USA
[7]Department of Neuroscience, Swartz Program in Theoretical Neuroscience, Kavli Institute for Brain Science, College of Physicians and Surgeons and Mortimer B. Zuckerman Mind Brain Behavior Institute, Columbia University, New York City, NY 10027, USA
[8]Department of Physics, Duke University, Durham, NC 27710, USA
[9]These authors contributed equally
[10]These authors contributed equally
[11]Lead contact
*Correspondence: ken@neurotheory.columbia.edu (K.D.M.), nicolas.brunel@duke.edu (N.B.)
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SUMMARY

The ability to optogenetically perturb neural circuits opens an unprecedented window into mechanisms governing circuit function. We analyzed and theoretically modeled neuronal responses to visual and optogenetic inputs in mouse and monkey V1. In both species, optogenetic stimulation of excitatory neurons strongly modulated the activity of single neurons yet had weak or no effects on the distribution of firing rates across the population. Thus, the optogenetic inputs reshuffled firing rates across the network. Key statistics of mouse and monkey responses lay on a continuum, with mice/monkeys occupying the low-/high-rate regions, respectively. We show that neuronal reshuffling emerges generically in randomly connected excitatory/inhibitory networks, provided the coupling strength (combination of recurrent coupling and external input) is sufficient that powerful inhibitory feedback cancels the mean optogenetic input. A more realistic model, distinguishing tuned visual vs. untuned optogenetic input in a structured network, reduces the coupling strength needed to explain reshuffling.

INTRODUCTION

Since their introduction nearly two decades ago, optogenetic methods have revolutionized neuroscience.[1] Optical stimulation of specific cell types during recordings of neuronal activity can probe the network operating regime, test predictions from theoretical models,[2–6] and provide insight into synaptic connectivity.[7–11]

Here, to probe the operating regime of the sensory cortex[12] and the mechanisms that shape its responses to its inputs, we analyze electrophysiological recordings from both mouse[13] and monkey[14] primary visual cortex (V1) during presentation of visual stimuli of various contrasts and optogenetic stimulation of excitatory (E) neurons. We find that optogenetic stimulation yields a broad distribution of firing-rate changes in single neurons, with some neurons strongly excited and others strongly suppressed. Yet, changes in mean population responses are much smaller than typical single-neuron firing-rate changes. Indeed, the mean and the entire distribution of firing rates are largely unaffected by optogenetic stimulation (significant changes seen only at low visual contrast in mice), even though the variance of firing-rate changes is comparable to the variance of rates prior to optogenetic stimulation. Thus, optogenetic stimulation of E neurons leads to a reshuffling of firing rates across the network, with little or no change in their distribution. We find a continuum from mice at low contrast to mice at high contrast to monkeys at low contrast to monkeys at high contrast, along which firing rates increase and the optogenetically induced changes in the mean and distribution of firing rates shrink to zero.

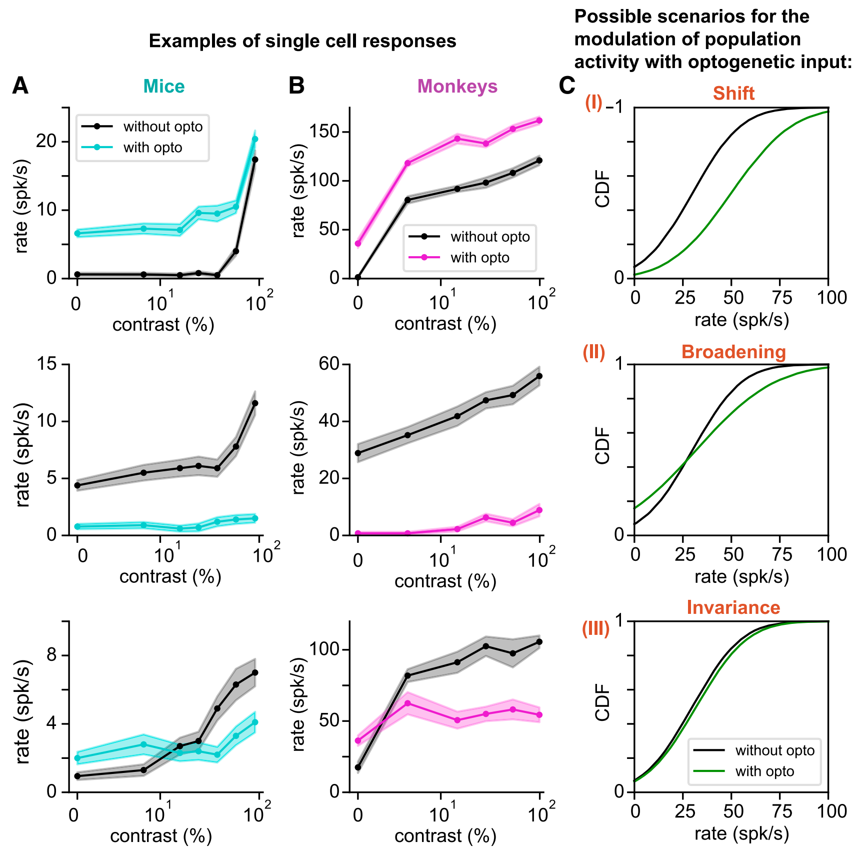

Figure 1. Optogenetic modulation of visual responses in mice and monkeys: single-neuron examples and potential population scenarios

(A) Examples of single-neuron responses to visual stimuli in mice, without (black) and with (colored) optogenetic stimulation of E neurons (here and in the following figures, lines represent means, shaded regions are mean ± SEM).

(B) As in (A) but in monkeys. In both species, optogenetic stimulation strongly increases the rates at all contrasts for some neurons (top row); others are strongly suppressed (center row); and others exhibit excitation at low contrast, but suppression at high contrast (bottom row). The same color code is used in all the following figures.

(C) Possible scenarios of how optogenetic stimulation of E neurons might modify the network's distribution of firing rates (see text).

We explore the mechanisms of such reshuffling using theoretical analysis and modeling. We first examine randomly connected networks of E neurons and inhibitory (I) neurons. We find that such networks reproduce all the above-mentioned features; provided the coupling—determined by a combination of the strengths of recurrent coupling and external input—is strong enough, the network is inhibition dominated, and opsin-induced currents are sufficiently heterogeneous and are sufficiently weak in the mean relative to visually induced currents. The continuity from mice to monkeys can be understood if mice are in a more weakly coupled regime than monkeys. However, for full reshuffling to arise in the randomly connected model, the required coupling is very strong—stronger than suggested by review of relevant experiments[12]—and the optogenetic input is much weaker than the visual input. Because it is unlikely that the first, and unclear whether the second, of these conditions are realized in mice or primate V1, we also investigate structured network models with connectivity dependent on spatial distance and orientation tuning, which can incorporate the difference between tuned visual and untuned optogenetic input. In the structured networks, reshuffling again requires sufficiently strong coupling and sufficiently weak optogenetic input but can arise with more biologically plausible parameters, i.e., weaker coupling and stronger optogenetic input. These results provide new insights into the operating regime of the sensory cortex and how this may differ across mammalian species.

RESULTS

Heterogeneous optogenetic modulation of visual responses in mice and monkeys

We analyzed recordings from the V1 of mice (data from Histed[13]) and monkeys (data from Nassi et al.[14]). These experiments shared a number of common features (see STAR Methods for details): visual stimuli were gratings of different contrasts; virally expressed opsins were used to optogenetically stimulate pyramidal cells. Both datasets consist of neural spiking responses recorded extracellularly in awake animals to combinations of visual and optogenetic stimuli. There are also a few differences between the experiments. In monkeys, visual stimuli were drifting sinusoidal gratings centered within the receptive field and matched to each recorded unit's preferred size, orientation, and spatial and temporal frequencies. In mice, multiple neurons were recorded simultaneously and there was no systematic relationship between the visual stimulus and the preferences of the recorded neurons.

In both mice and monkeys, as previously observed,[13,14] single-neuron responses to visual stimuli were strongly modulated by optogenetic stimuli, in a highly heterogeneous fashion (Figures 1A and 1B). Responses to visual stimuli were typically either enhanced by the laser at all contrasts, suppressed by the laser at all contrasts, or enhanced at low contrast and suppressed at high contrast (Figures 1A and 1B); neurons suppressed at low contrast and enhanced at high contrast were not observed. These single-neuron responses might reflect different scenarios at the population level (Figure 1C). Optogenetic stimulation might: (1) significantly increase the mean activity in the network, while broadening the distribution of single-neuron rates. This scenario is what one would expect naively from optogenetic stimulation of E neurons. (2) Broaden the distribution of rates, but with a weak mean increase of the population activity due to many suppressed cells. (3) Only weakly affect the

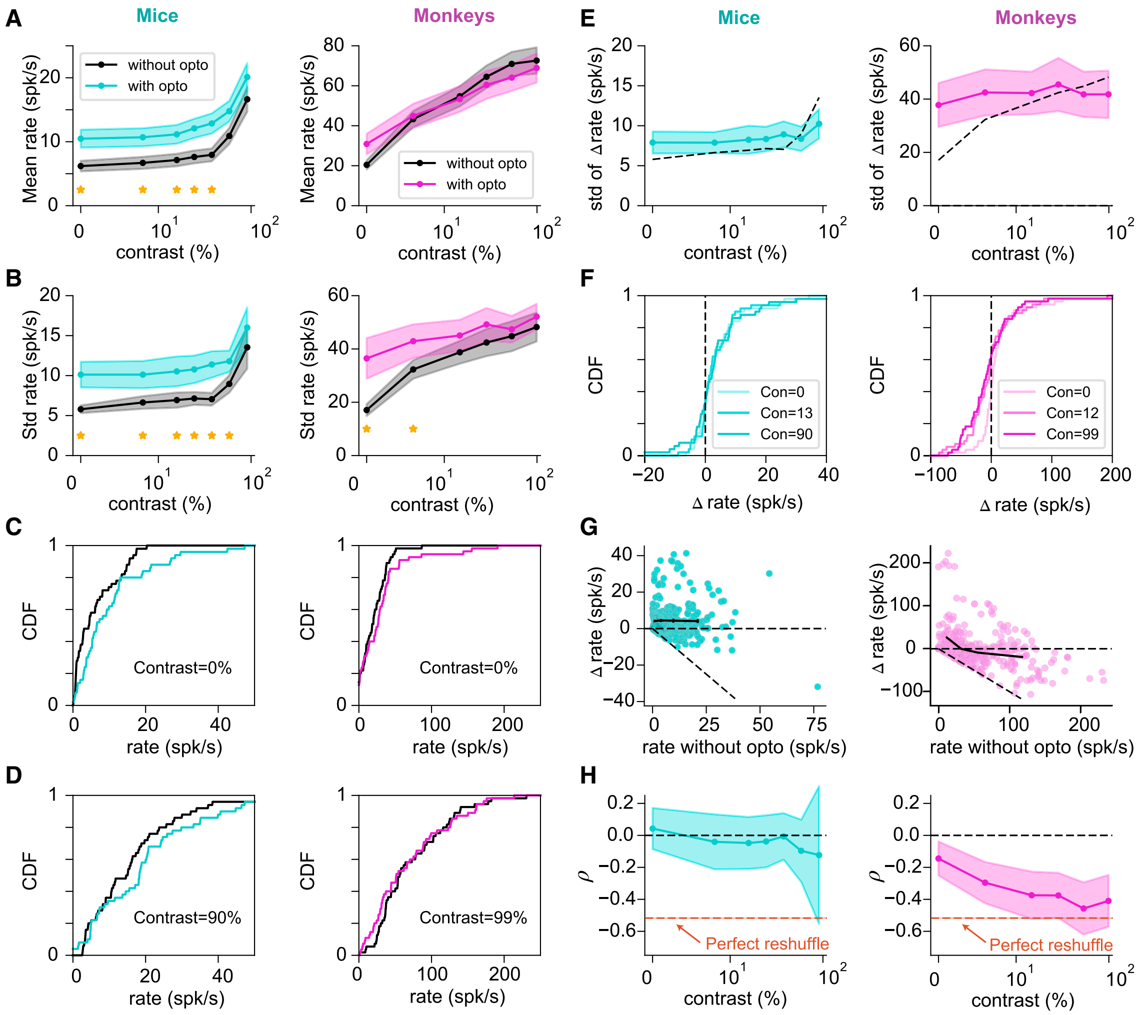

Figure 2. Effects of optogenetic stimulation on population statistics in mice and monkeys

(A) Population mean response to visual and optogenetic stimuli. Optogenetic modulation caused no significant change in mean rates except in mice at visual stimulus contrast <55% (stars).

(B) Standard deviation of the distribution of firing rates is increased significantly by light only for contrasts <90% in mice and <12% in monkeys.

(C and D) Cumulative distribution function (CDF) of firing rates at 0% (C) and high (D) contrasts. Optogenetic modulation significantly changed the CDF only for mice at low contrast (0%, 6%, and 12%).

(E) In both species, the standard deviation of the optogenetically induced change in firing rate (shown as a function of contrast, colored lines) was larger than the mean change and of the same order as the standard deviation of rates without the laser (dashed black lines, same as black lines in B).

(F) Cumulative distribution function of optogenetically induced changes in firing rate (Δr) at three example contrasts (indicated by transparency levels). In both species, and at all contrasts, Δr was broadly distributed, with a large fraction of suppressed cells.

(G) Δr vs. baseline activity (i.e., activity without optogenetic stimuli). Each cell is represented once for every value of stimulus contrast. Continuous black lines represent mean Δr in quartiles of the distribution of baseline rates.

(H) Normalized covariance ρ between Δr and baseline rates ($\rho = \mathrm{Cov}(r, \Delta r)/\mathrm{Var}(\Delta r)$). In mice, ρ is not significantly different from zero. In monkeys, ρ decreases with contrast, approaching -0.5, which, as described in the text, is the value required for perfect invariance of the rate distribution.

distribution of rates despite strong changes in individual neurons. Which of these scenarios better represents the effect of optogenetic stimulation? Do these effects change with contrast and species?

Optogenetic stimulation reshuffles visual responses in mice and monkeys

To answer the above-mentioned questions, we analyzed the statistics of responses to visual and optogenetic stimuli (Figure 2). In both mice and monkeys, mean population activity increased monotonically with visual stimulus contrast, but mice had significantly lower mean firing rates (ranging from 6 ± 0.9 spikes/second (spk/s) [mean ± SEM] at 0% contrast to 16.7 ± 2.0 spk/s at high contrast) than monkeys (ranging from 20.2 ± 2.0 spk/s to 72.7 ± 6.7 spk/s) (Figure 2A). In both species, visual stimuli of fixed contrast generated a broad distribution of rates (Figure 2B), and increasing contrast significantly shifted distributions toward higher rates (Figures 2C and 2D). Surprisingly,

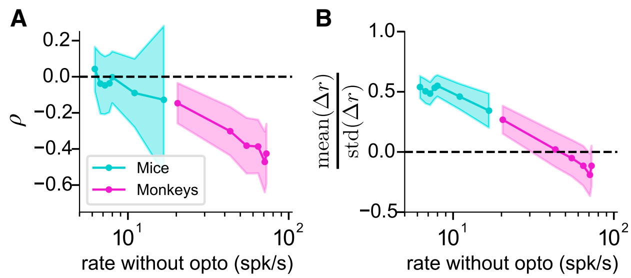

Figure 3. Optogenetic responses in mice and monkeys as a function of average visual response

(A) Normalized covariance (Equation 1) and (B) ratio of mean over standard deviation of the change in rate as a function of average visual response (different points correspond to different contrasts). Data across the two species trace out a single curve.

optogenetic stimulation had much weaker effects at the population level (Figures 2A–2D). In mice, the mean and standard deviation of the population activity were significantly modulated by the optogenetic stimulation only for contrasts lower than 55% (Welch's t test, $p < 0.05$) and 90% (f test, $p < 0.05$), respectively. In monkeys, the mean population activity was not significantly modulated at any contrast (Welch's t test, $p > 0.05$), while the standard deviation was significantly modulated only at contrasts lower than 12% (f test, $p < 0.05$). Even more strikingly, optogenetic stimuli weakly affected the whole distribution of rates in the network, with no significant changes in distribution according to a Kolmogorov-Smirnov (KS) test ($p > 0.05$) except at low contrasts in mice (Figure 2D). (Note that significant changes in mean or standard deviation imply changes in distribution, but some of these changes were not detected by the KS test.)

Though we observe weak optogenetic modulation at the population level, the distributions of optogenetic modulations of single-neuron firing rates were broad in both species, with standard deviation comparable to or larger than the standard deviation of the rates without the optogenetic stimulus (Figures 2E and 2F). Furthermore, unlike visually induced responses (Figures 2C and 2D), the optogenetically induced rate changes had means considerably smaller than their standard deviations (Figures 2A and 2E): combining all contrasts, the mean ± SD of optogenetically induced rate changes was 4.2 ± 8.9 spk/s in mice and -0.7 ± 43 spk/s in monkeys. The distribution of optogenetic responses showed no significant differences between contrasts in mice, while in monkeys, the only significant difference was between zero contrast and all other contrasts (KS test, significance assayed as $p < 0.05$). In both species, a significant fraction of cells were suppressed (e.g., at high contrast, $16/55$ in mice and $37/50$ in monkeys, respectively). In sum, optogenetic stimuli *reshuffled* visual responses in mice and monkeys: they strongly modulated single-neuron responses while causing only a weak modulation of population activity.

How can optogenetic stimulation leave the distribution of rates unchanged, while at the same time yield large changes in firing rates of individual neurons (Figure 2B)? If optogenetic responses were independent of baseline firing, as one might expect intuitively, optogenetic stimuli would significantly broaden the distribution of visual responses. The rate variance with ($\sigma^2_{r+\Delta r}$) or without (σ^2_r) optogenetic stimuli are related by $\sigma^2_{r+\Delta r} = \sigma^2_r + \sigma^2_{\Delta r} + 2\mathrm{Cov}(r, \Delta r)$ (where "Cov" is covariance). Thus, independence of r and Δr would imply the rate variance is increased by $\sigma^2_{\Delta r}$, which, as we have seen (Figure 2E), is comparable in size to σ^2_r. If the distribution of rates is not modified by optogenetic stimuli, then $\sigma^2_{r+\Delta r} = \sigma^2_r$, which implies a negative covariance, $\mathrm{Cov}(r, \Delta r) = -0.5\sigma^2_{\Delta r}$.

In mice, the change in rate induced by laser Δr and rate without laser r are only weakly anticorrelated (Figure 2G, left; Pearson correlation coefficient -0.06, combining all contrasts), consistent with a significant optogenetically induced increase in the width of the rate distribution. In monkeys, Δr and r are strongly anticorrelated (Figure 2G, right; Pearson correlation coefficient -0.38, combining all contrasts). That is, neurons with higher (lower) baseline rates were more likely to be suppressed (excited) by optogenetic stimuli. To quantify the degree to which this anticorrelation is consistent with an invariant firing-rate distribution, we define a normalized covariance

$$\rho = \frac{\mathrm{Cov}(r, \Delta r)}{\sigma^2_{\Delta r}}. \qquad \text{(Equation 1)}$$

ρ is equal to -0.5 when optogenetic stimuli leave the rate distribution invariant. In mice, ρ is never significantly different from zero (Figure 2H, left). In monkeys, ρ is significantly negative at all contrasts (Welch's t test, $p < 0.05$) and close to -0.5 at high contrast (Figure 2H, right), consistent with invariance of the rate distribution, i.e., with reshuffling of rates by optogenetic stimulation.

Are quantitative species differences in optogenetic responses related to differences in baseline firing rates? To test this, we examined dimensionless variables characterizing the effects of optogenetic stimuli on the statistics of network activity for a given visual stimulus (including a 0% contrast stimulus) as a function of the average firing rate induced by that visual stimulus. Remarkably, curves traced by monkey data seem to be the continuation of those for mouse data, with mice occupying the low-rate region and monkeys the high-rate region of underlying common curves (Figures 3A and 3B). In particular, in both species, with increasing visually induced firing rates, the mean of Δr relative to its standard deviation decreases, approaching zero at a high firing rate, and the normalized covariance ρ also decreases, approaching -0.5 at a high rate.

Network reshuffling emerges in strongly coupled random network models

To understand the mechanisms underlying reshuffling of visual responses by optogenetic stimuli, as well as the observed differences between mice and monkeys, we first investigated the response properties of a simple model of cortical circuits, a network of randomly connected E- and I-rate neurons[15–17] (from now on: "unstructured model," Figure 4A). The single-neuron response function (firing rate vs. input current) was taken to be the f-I curve of leaky integrate-and-fire neurons driven by white noise,[18–20] which is supralinear at lower mean input and becomes sublinear at high input due to saturation (results presented here do not depend on this specific choice, Figure S1). Visual stimuli are represented by external inputs to E and I units, while optogenetic inputs are an additional input to E cells.

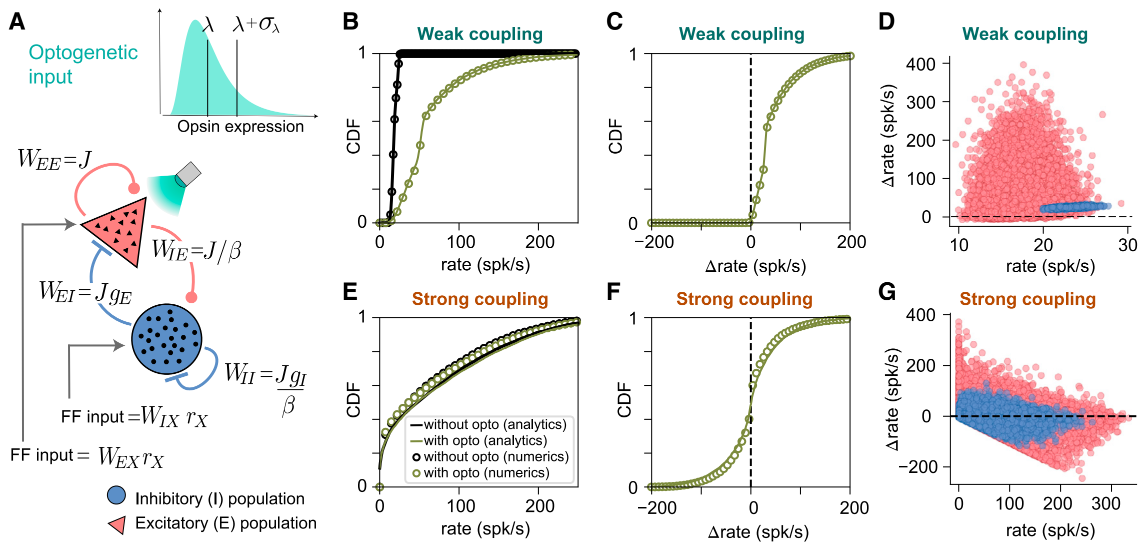

Figure 4. Strong coupling leads to large fractions of suppressed cells, negative covariance, and activity reshuffling in randomly connected E-I network models

(A) Schematic of unstructured network model, with randomly connected E- and I-rate units. The network is driven by a feedforward (FF) visual input, targeting both cell types, and by an optogenetic input targeting only E neurons. Visual inputs follow a Gaussian distribution, while opsin expression follows a lognormal distribution.

(B–D) Responses of all cells to optogenetic stimulation of E cells in a weakly coupled network model. In all panels, dots represent numerical simulations while lines are mean field theory predictions. For weak coupling, optogenetic stimulation strongly affects the distribution of firing rates (B), with only a weak fraction of suppressed cells (C). Changes in firing rates are positively correlated with baseline activity (D).

(E–G) As in (B)–(D), but in a strongly coupled network. For strong enough coupling, optogenetic stimulation does not significantly affect the distribution of rates (E). Firing-rate changes are broadly distributed, with a mean around zero and a large fraction of suppressed cells (F). Firing-rate changes are anticorrelated with baseline activity (G). Parameters in table of parameters of unstructured models, STAR Methods.

See also Figures S1–S5.

Through numerical simulations and analytical calculations, we found that this model reproduces the main features of the experimental data, provided that recurrent connectivity and visual input are strong enough and the distribution of opsin expression is sufficiently heterogeneous and weak. We first focus on the dependence of optogenetic responses on the network's coupling strength, which can be thought of as the change in one neuron's rate per change in another's rate. This is the product of the synaptic strength and the sensitivity (or gain) of the postsynaptic cell to synaptic input,[12] i.e., the slope of its input/output function. In the supralinear portion of the input/output function, this increases with network activation and thus with the strength of visual input.

In the absence of optogenetic stimulation, and consistent with previous theoretical studies,[20–23] the distribution of rates is narrow for weak coupling and broad for strong coupling (Figures 4B and 4E). Responses to optogenetic stimulation strongly depend on the strength of coupling. For weak coupling, optogenetic stimulation significantly shifts the distribution of rates (Figure 4B). This leads to a broad distribution of optogenetic responses, characterized by a positive mean and a lack of significantly suppressed cells (Figure 4C). Single-neuron responses to optogenetic stimulation are positively correlated with baseline activity (Figure 4D, Pearson correlation coefficients: 0.1 [E cells], 0.6 [I cells]). This positive correlation arises because the supralinear input/output function (which is stronger for I cells because of their shorter membrane time constant[24]) makes isolated neurons more sensitive to inputs at higher firing rates, and weak coupling renders network effects weak. The responses in weakly coupled networks match naive expectations but strongly differ from experimental observations. When coupling is strong, on the other hand, optogenetic stimulation does not significantly modify the distribution of rates (Figure 4E). It produces a broad distribution of rate changes that is negatively correlated with baseline activity (Figure 4G), with a mean close to zero (Figure 4F). These response properties are strikingly similar to experimental observations.

Mechanisms shaping the response to optogenetic inputs in random network models

We performed numerical and mean field analyses of the network (details in STAR Methods) to elucidate the circuit mechanisms underlying the simulation results of Figure 4. These analyses show that, with strong coupling, optogenetic input evokes only a weak change in average firing rate, provided the network is inhibition dominated (Figure S2). In particular, we find that the average rate change due to optogenetic stimulation decreases when: (1) coupling strength is increased, by increasing either the firing rate r_X of the network's visually driven inputs or J, which scales both recurrent and visually driven synaptic weights (Figures S2A–S2C, first row); or (2) inhibition becomes more dominant, either by increasing strength of inhibition onto E neurons, g_E, or decreasing strength of inhibition onto I neurons, g_I.

These results can be understood analytically from the implicit equations defining the mean rates (Equation 15, STAR Methods). We linearize the equations about the fixed point by assuming optogenetically induced mean rate changes, Δr_E and Δr_I, are small. In the limit of small mean optogenetic input λ, and neglecting contributions from randomness in the connectivity and in opsin expression, we can solve these equations to find

$$\Delta r_E = \frac{g_I \lambda}{KJ\tau_E(g_E - g_I)}, \quad \Delta r_I = \frac{\lambda}{\gamma KJ\tau_E(g_E - g_I)} \qquad \text{(Equation 2)}$$

where K and γK are the average number of E and I recurrent connections per neuron, while τ_E is the membrane time constant of E cells. Equation 2 qualitatively captures the dependence on J and $g_{E,I}$ observed in simulations (Figures S2A–S2C, first row).

Strong recurrent input can largely cancel the mean optogenetic input, but cannot cancel the cell-to-cell fluctuations in input induced by opsin expression heterogeneity.[25,26] This produces high response heterogeneity (i.e., high standard deviation of rate changes, Figure S2, second row). This heterogeneity increases with opsin expression variability (σ_λ, Figure S2D, second row) and only weakly depends on coupling strength (Figures S2A–S2C, second row).

Increasing coupling strength and heterogeneity of opsin expression leads the normalized covariance ρ to become negative and close to -0.5 (Figure S2, third row; Pearson correlation coefficients: -0.3 [E cells], -0.2 [I cells]) and yields network reshuffling (Figures 4E–4G). ρ becomes negative because of an interplay between single-neuron nonlinearities and strong I feedback. Strong couplings yield a broad distribution of baseline rates and, therefore, a large fraction of cells with very low baseline rates. Upon heterogeneous optogenetic stimulation, the non-negativity of firing rates, and more generally the expansive, supralinear neuronal input/output function at low firing rates, biases the responses of cells with low baseline rates toward excitation (see Figure 4G). To cancel the mean change in rate, the circuit must then, on average, suppress neurons with higher baseline activity (see Figure 4G). This yields the negative correlation between baseline activity and optogenetic response.

The dynamics of rate networks become chaotic when recurrent connections are strong.[15–17] Consistent with this, chaotic dynamics emerged in our simulations as J and/or r_X increased (Figure S2, lines under first row). We wondered whether the appearance of reshuffling is related to chaotic dynamics, but found that the parameter regions where these two phenomena occur only partially overlap (see Figure S3). That is, negative covariance and chaotic dynamics are two distinct phenomena that both emerge in networks of strongly coupled neurons.

Randomly connected network models capture experimental recordings in mice and monkeys

We next examined whether, and under what conditions, unstructured models could quantitatively fit the data from the two species. We saw previously that key population response statistics seemed to follow a single curve vs. baseline firing rate (Figures 3A and 3B). This, in turn, might be due to a difference in visual input strengths in the two species. This motivates the hypothesis that the strength of external inputs might be the main factor explaining differences between mice and monkeys.

To test this hypothesis, we fit a single unstructured network model to the data from both species, with only the strengths of visual (r_X) and optogenetic (L) inputs differing between them. Our best-fit such model (see STAR Methods) captures the main features of the recordings (Figures 5A and 5B, dashed lines): weak mean but large standard deviation of optogenetic response and increasingly negative normalized covariance with increasing visual stimulus contrast. It also captures key differences between mice and monkeys: a lower visual response and a larger mean to standard deviation ratio of the optogenetic response in mice and a more negative normalized covariance in monkeys. These results support the idea that the strength of external inputs is a key factor shaping differences between the two species.

At the same time, the model fails to capture quantitatively the mean optogenetic response and the amplitude of the negative normalized covariance in monkeys. These discrepancies suggest that differences in the network connectivity, or the visual or optogenetic input distributions (i.e., their variances), might be important to fully capture the observed response properties. To test this, we repeated the fitting procedure in unstructured models, allowing all parameters to differ between mice and monkeys. We found that this model gives a better description of experimental recordings, including the strongly negative covariance seen in monkeys for high contrast (Figures 5A and 5B, continuous lines; relative error, averaged over the quantities plotted in Figures 5A and 5B, decreased by 22% in mice and 5% in monkeys). For the best-fit parameters (table of parameters of unstructured models, STAR Methods), mice have weaker overall synaptic coupling J, weaker inhibition (weaker overall $I \to E$ strength, Jg_E, and ratio of $I \to E$ to $I \to I$ strength, g_E/g_I), and less heterogeneous optogenetic expression (smaller σ_λ/λ) than monkeys.

The best-fit parameters frequently fell within the chaotic regime (Figure 5A) (see STAR Methods). In the simultaneous (independent) fits, chaos arose in mice at intermediate (intermediate to high) contrasts and in monkeys at all (low) contrasts.

Previous theoretical work[4,12,27,28] suggested that cortical circuits operate in a loosely balanced regime, characterized by the net input to cells (E minus I) being comparable to the individual (E or I) components. This can be quantified by the *balance index*, the ratio of the net input to the total E input; in general, tighter balance (smaller balance index) corresponds to stronger coupling.[12] Using the best-fit model parameters inferred from data, we found a smaller balance index in monkeys than in mice (Figure 5C). In mice, the network was loosely balanced (balance index > 0.1) at low and intermediate contrast. When mice and monkeys were jointly fit, somewhat tighter balance was observed at high contrast in mice and for all contrasts in monkeys (Figure 5C, dashed line), while monkeys showed very tight balance when independently fit (Figure 5C, filled line).

The unstructured model also requires the ratio of mean optogenetic input to visually evoked E input to be small (~ 0.1) for the jointly fit model and the independent fit to mice, and very small ($\lesssim 0.01$) for the independent fit to monkeys (Figure 5D). The latter

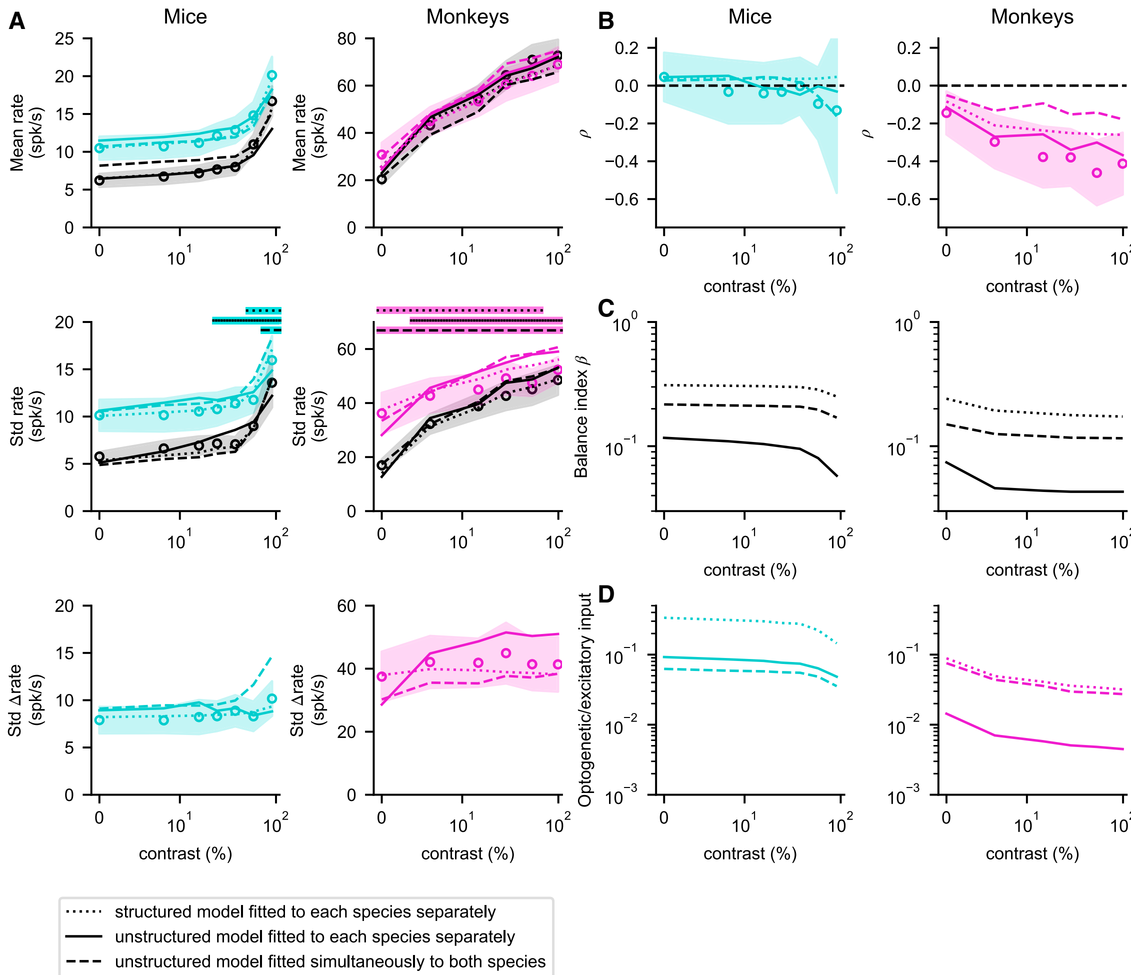

Figure 5. Fitting mice and monkey data with the unstructured, randomly connected, and structured network models

(A and B) Comparison between data (circles, shaded regions show confidence region; see STAR Methods) and network models with parameters that best-fit data in mice (left) and monkeys (right). We show best fits of the unstructured model, either using the same parameters in mice and monkeys, except for different inputs (dashed lines), or using different model parameters in the two species (continuous lines). Dotted lines represent best fits of the structured model with different model parameters in the two species. Horizontal bars in the second row of (A) indicate contrasts at which each model has weakly chaotic dynamics (see STAR Methods). Model plots are from simulations, averaging over 20 realizations. Network models reproduce key features of experimental data, including weak mean changes in firing rate and broad distribution of responses (A), and increasingly negative covariance with contrast of the visual stimulus (B).

(C) Best-fit model values of the balance index.[12] Stronger coupling yields tighter balance, i.e., a smaller balance index. Structured model fits are less strongly coupled (larger balance index) than unstructured model fits.

(D) The ratio between optogenetic input and the total visually driven excitatory input in the best-fit models is larger in structured models than in unstructured models.

See also Figure S6.

is related to tight balancing. In the independent fit to monkeys, the mean I recurrent input cancels most of the mean E input at higher contrasts (balance index $\approx$0.04, Figure 5C). Both the visual and the optogenetic input are strongly cancelled, by similar factors. For the mean optogenetic response to be reduced to near zero by cancellation, while the mean visual response remains finite, the mean optogenetic input must be much smaller than the mean visual input. Thus, by both measures—the balance index and the relative mean size of optogenetic vs. visual input—the nearly perfect reshuffling seen in monkeys at higher contrast appears to require tight balancing (very strong coupling), tighter than appears to be the case for the V1 or other cortical areas from previous reviews of the relevant literature.[12]

Structured connectivity leads to reshuffling with weaker coupling

The unstructured models studied to this point ignored an important difference between the visual and optogenetic input: the visual input is *tuned*, specifically targeting a set of cells with similar feature and spatial selectivity, while the optogenetic input is untuned, targeting cells regardless of their selectivity. They similarly ignored the tuning of connectivity, which decreases with cortical distance[29,30] and with difference in feature preference.[10,31,32] We reasoned that a tuned stimulus would excite cells with matched tuning, which might preferentially excite one another and/or inhibit others, whereas an untuned stimulus would tend to excite and inhibit all cells equally, which could give a very small or even negative response. With such a mechanism, the network might not need such tight balance or small optogenetic input to explain reshuffling.

To test these ideas, we constructed structured models. We assumed that neuronal selectivity was determined by a single feature, preferred orientation. Connection strengths decayed as a product of Gaussian functions of the physical distance between neurons and of the difference in their preferred orientations. We used a spatially salt and pepper organization of preferred orientation in the mouse and a quasi-periodic map in the monkey, as reported experimentally[33–36] (Figure S4). Visual input was also a product of Gaussian functions of the distance between receptive field center and stimulus location and of the difference between preferred and stimulus orientations. The experimental recordings matched visual stimuli to cell preferences in monkeys[14] but not in mice.[13] Accordingly, we fit the response statistics of the stimulus-matched cells (those with preferred orientation and retinotopic position matching the visual stimulus) in the monkey models, and of all cells in the mouse models (STAR Methods).

We found that, for both mice and monkeys, the same quality of fit could be obtained with the structured models as with the unstructured models (Figures 5A and 5B). However, compared with the independently fit unstructured model, the structured model matches the monkey's nearly perfect reshuffling with a much larger balance index ($\sim$ 5 times larger, Figure 5C) and a much larger ratio of mean optogenetic to visual input ($\sim$ 5 times larger, Figure 5D).

Although the structured models can fit the data with weaker coupling (larger balance index), both structured and unstructured models require sufficiently strong coupling to capture the weak modulation of the means and distributions of rates (Figures 4 and S4–S6). Both unstructured and structured models showed qualitatively similar dependence on parameters, including J and r_X, either of whose increasing strength represents increasingly strong coupling (Figures S2 and S5). Both models required large J and/or r_X to exhibit reshuffling, and in both, increasing J or r_X tended to generate more negative mean rate changes and covariances.

To better characterize the coupling strength required to fit the data, we studied the effects of varying J in the monkey models for a high-contrast visual stimulus, the case with the purest reshuffling. For each value of J, we repeated the fitting procedure, finding the best-fitting model for the given J. For both structured and unstructured models, the balance index decreases with increasing J (Figure S6C), consistent with our identification of stronger coupling (larger J) with tighter balance. Both structured and unstructured models reproduced the neuronal reshuffling measured in monkeys ($\rho \sim -0.5$) only for sufficiently large coupling and failed to do so at weak coupling (Figure S6A). Response properties for weak coupling are similar for the structured and unstructured models: the normalized covariance ρ is close to zero (Figure S6A) and the mean optogenetic response Δr is large and positive (Figure S6B). In sum, while the use of a structured rather than a random network quantitatively decreases the strength of coupling required to fit the data (i.e., allows fitting with larger balance index, Figure S6C, and larger ratio of optogenetic to total visually driven E input, Figure S6D), it does not modify the qualitative requirement of sufficiently strong coupling to explain strong reshuffling.

Finally, we analyzed the range of contrasts over which weak chaotic dynamics emerges in the structured model. Similar to the unstructured model, weak chaotic dynamics arose in mice at high contrast and in monkeys at low contrast (Figure 5A).

Network models reproduce single-neuron properties observed in data

Was fitting the models to the population statistics of Figure 5 sufficient to also lead them to reproduce the full distributions of single-neuron responses? In the unstructured and structured models and in the data: (1) optogenetic responses (Δr) were broadly distributed, with a considerable proportion of suppressed cells (Figure 6); (2) the Δr distribution varied little with contrast, except that monkeys and the unstructured monkey model showed weaker suppression at zero contrast (Figure 6, first and third columns); and (3) scatter plots of r vs. Δr are remarkably similar (Figure 6, second and fourth column). Thus, the model distributions of optogenetic responses are in good agreement with the data.

We next analyzed how optogenetic stimuli modulate the contrast response of single neurons. Previous work[14] showed that single-neuron responses to visual and optogenetic stimuli in monkeys can be described by a "normalization equation":

$$r(c,\lambda) = R_m(R_0 + c^n + \lambda D)\,/\,(\sigma + c^n + \lambda S) \qquad \text{(Equation 3)}$$

Here, c is the contrast of the visual stimulus, λ is 1 if the optogenetic stimulus is present and 0 otherwise, and the parameters R_m, R_0, n, σ, D, and S are fitted for each neuron. Equation 3 provides a concise description of single-neuron response and can be used to compare model results to experimental data. To this end, we fitted the normalization model to responses in the data and in network model simulations (examples in Figures 7A and 7B).

The normalization model captured much of the variance in single-cell responses, both in the data and in the models, capturing similar amounts of variance in the data and the unstructured model and even more variance in the structured model (Figure 7C). The D/S ratio (where D and S describe optogenetic response facilitation and suppression, respectively; Equation 3 favored S in the majority of cells in both data and models, with a similar distribution of ratios in data and the unstructured model and S less strongly favored in the structured model. Other normalization model parameters also show similar distributions in data and models, with the unstructured model giving a closer match to data than the structured model (Figure S7). Thus, our models, only constrained to reproduce first- and second-order population statistics, produce single-neuron contrast and optogenetic responses that reasonably resemble those seen in the experimental data.

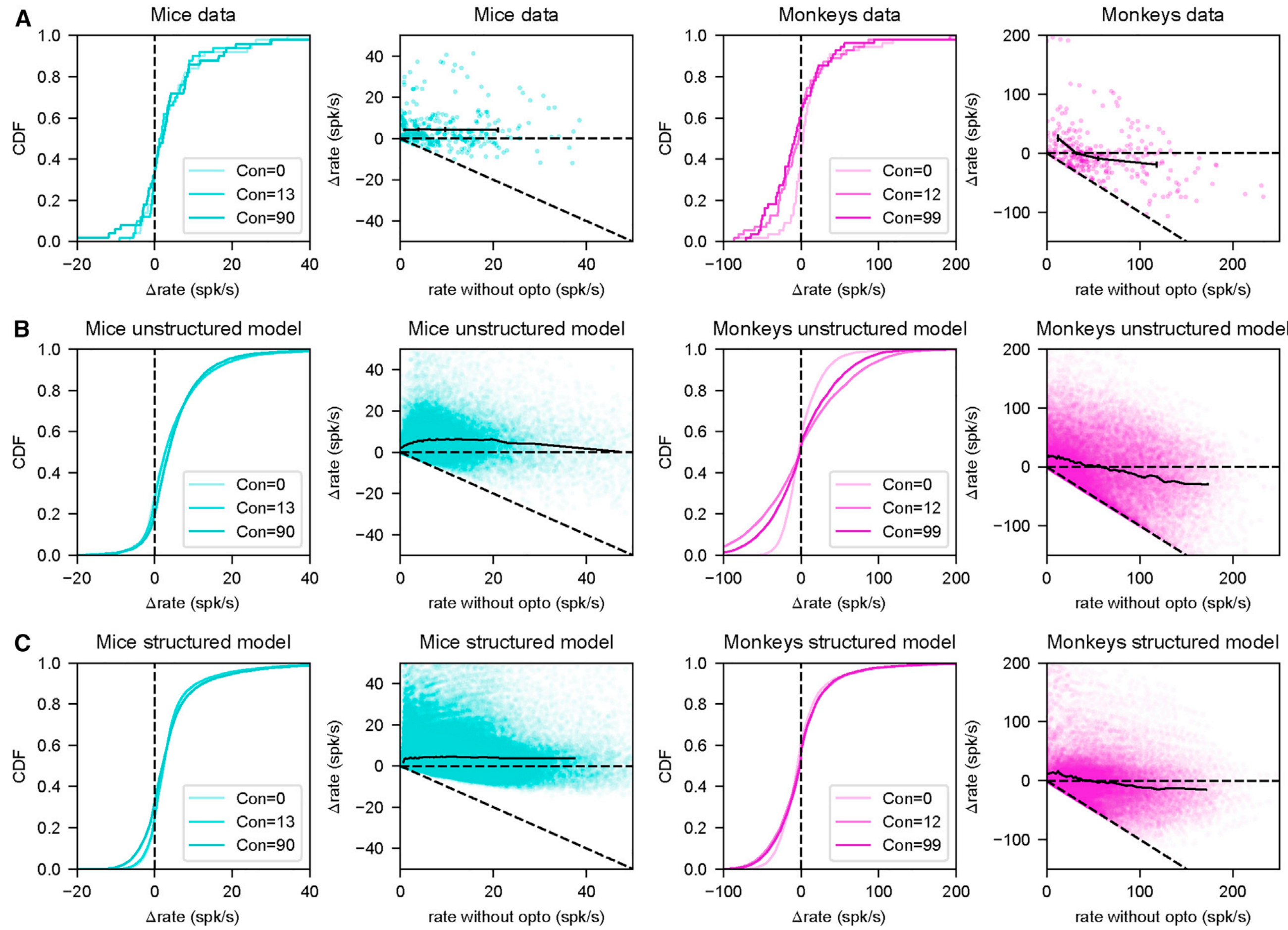

Figure 6. Network models recapitulate the heterogeneity of optogenetic responses measured in mice and monkeys

(A) Cumulative distribution of single-neuron optogenetic responses (Δ rate) for three contrasts, and Δ rate vs. baseline activity across all contrasts (each cell is represented once for each value of contrast), in mice (left) and monkeys (right); same data as Figures 2F and 2G.

(B) As in (A) but for the best independently fit unstructured models of Figure 5.

(C) As in (B) but for the best-fit structured models of Figure 5. Note the good agreement between data and models in all cases.

DISCUSSION

Summary of the results

We analyzed electrophysiological recordings of mice and monkey V1 during visual stimulation, with or without concurrent optogenetic stimulation of E neurons. In both species, and for all stimulus contrasts, optogenetic stimuli generated a broad distribution of responses, with mean considerably smaller than the standard deviation. The standard deviation is large, similar to that of visual responses. Despite this strong optogenetic modulation of the activity of single neurons, the distribution of rates across the population is not significantly changed by optogenetic stimulation in monkeys and, for high visual contrast, mice. This "reshuffling" of rates is specific to optogenetic stimuli; visual stimuli shift the distribution toward higher rates. Intriguingly, statistics of optogenetic responses across the two species form a single, continuous function of baseline (visually induced) population activity, with mouse and monkey responses corresponding to lower and higher baseline activities, respectively. Across the two species, increasing baseline activity yields increasingly strong reshuffling (optogenetically induced change in the response distribution's mean and standard deviation approaching zero, while the standard deviation of optogenetic responses remains large).

Why does optogenetic stimulation drive large single-cell responses but little or no mean response? To address this, we first studied an unstructured, randomly connected network model of E and I neurons. We found that the "balancing" that occurs in sufficiently strongly coupled E/I networks[12,25]—the dynamic cancellation of external input by recurrent input—can produce this effect. Balancing suppresses the mean input, but not the variation in input across cells. Accordingly, when the heterogeneity of optogenetic input (i.e., of opsin expression) is comparable to or larger than the mean optogenetic input, the mean response is suppressed while individual cells show large responses. However, the randomly connected model required much tighter balancing (more precise input cancellation, which arises with stronger recurrent coupling) than experiments

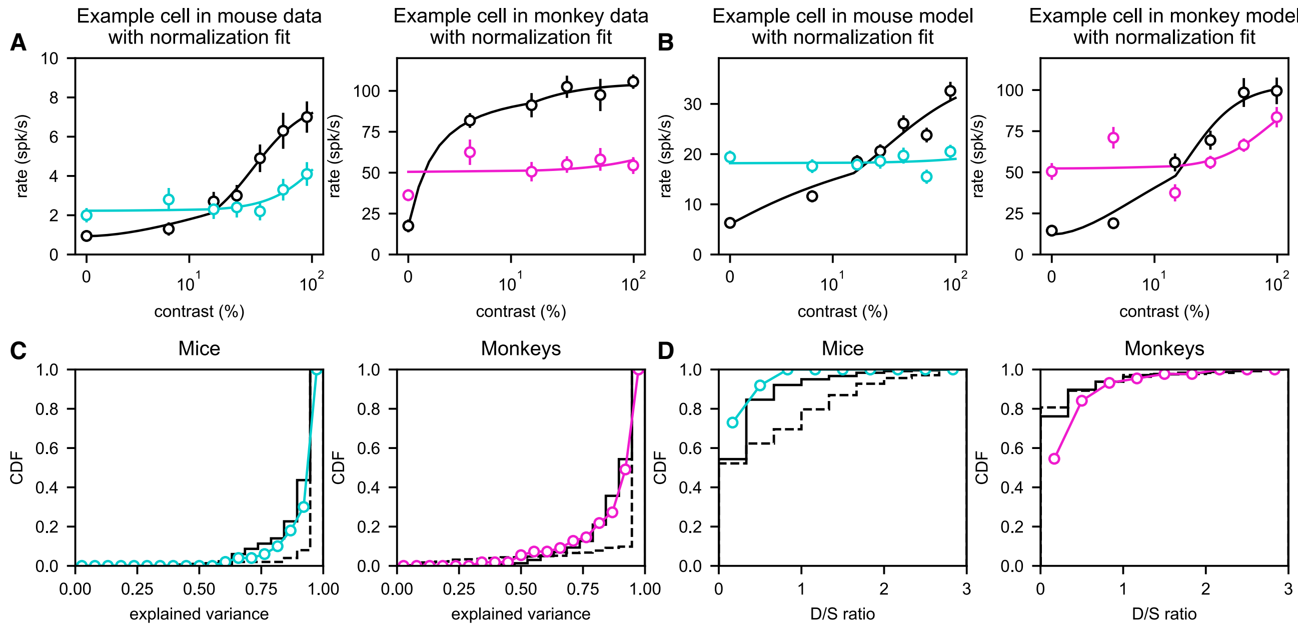

Figure 7. Fits of data and network simulations using the normalization model

(A) Example single-neuron responses from mice and monkeys, as in Figure 1. Measured responses (circles) can be captured by the normalization model (Equation 3; lines).

(B) As in (A) but for example neurons from the unstructured model with best independently fit parameters of Figure 5. In (A) and (B), we selected neurons with qualitatively similar behaviors: excited by light at low contrast, suppressed at high contrast.

(C) Cumulative distribution function of explained variance R^2 of the normalization equation fits in the data (colored lines) and in network models (continuous and dashed black lines represent best-fit unstructured and structured models of Figure 5, respectively).

(D) Cumulative distribution function of the D/S ratio, see Equation 3, which quantifies the ratio between the excitatory drive and divisive suppression induced by optogenetic stimuli, in data and models (same colors/symbols as in C).

See also Figure S7.

suggest.[12] It also required very small mean optogenetic input relative to visual input, so that suppression of the mean by balancing could produce almost zero mean optogenetic response with reasonable visual response.

We then incorporated the fact that the visual input is tuned, e.g., it targets cells with preferred orientation near the stimulus orientation, while the optogenetic input targets all E cells. We reasoned that a tuned input might lead to stronger excitation and/or weaker inhibition than an untuned input,[37] allowing suppression of the mean response under less stringent conditions. In the randomly connected model, optogenetic input was distinguished from visual only in projecting solely to E and having much smaller mean and larger variability relative to mean. Hence, we turned to a model with structured connectivity (feature- and distance-dependent) and tuned visual input. We found that it could achieve equally good fits to the data as the unstructured model, but with looser balancing in the reshuffled cells and with larger optogenetic input relative to visual. Both the unstructured and structured models require sufficiently strong coupling to fit the data, although the strength required for the structured model is less.

Theoretical implications: Coupling strength in cortex

Balanced network models reproduce many features of cortical activity.[12,20–28,38–44] Here, we showed that balanced networks can reproduce the surprising effect of strong reshuffling, adding to evidence that recurrent connectivity in the V1 is powerful and stabilized by strong inhibition.[2–4] In the balanced regime, the neuronal input/output function is generally supralinear,[45,46] so that increasing activation yields increasingly strong coupling.[12,27,28] In our models, this accounts for the strong dependence of the results of optogenetic stimulation on the state of the network prior to stimulation, as determined by the visual stimulus contrast and the species.

Balance may be loose[12,27,28] or tight[21,25,26] (balance index $\gtrsim$0.1 or $\lesssim$0.1, respectively), with tighter balance corresponding to stronger coupling. The unstructured model illustrates basic mechanisms but requires relatively tight balance. The structured model shows that similar mechanisms can operate with loose balance, consistent with estimates from experimental data.[12] However, those estimates were based on cat and mouse V1; it is possible that the monkey V1 has tighter balance, e.g., due to its higher firing rates (although extracellular recording may overestimate firing rates[47]).

An interesting feature of our analysis is that reshuffling and chaos often coexist.[15–17,48] However, our results do not provide strong evidence of chaos in cortical circuits. Although we found that weakly chaotic networks often provide better fits, non-chaotic networks can replicate the main features observed in the data.

Mice vs. monkeys

Our work provides, to our knowledge, the first systematic comparison of how optogenetic stimuli modulate visual responses in mice and monkeys. It shows that baseline firing plays a key role in determining the statistics of neural responses and suggests that the different activity levels, likely due at least in part to different strengths of external input, could underlie differences observed in the two species. When key statistics were plotted as a function of baseline rate, curves traced by mice and monkey data seem to lie on the same underlying curve, with mice occupying the low-rate region while monkeys occupy the high-rate region. Consistent with this observation, a single network model, with differing input strengths, gives a reasonable fit to the data of both species, though the quality of the fit improves when network parameters are allowed to differ between the two species.

A recent electron microscopy study performed a detailed comparison of reconstructed neurons and their synaptic connections in mouse and primate V1.[49] Surprisingly, synaptic connectivity in primates was much sparser than in mice, with a smaller ratio of E to I synapses than in mice. The larger ratio of E to I synapses in mice seems consistent with the weaker inhibition we found in mice. However, the weaker anatomical coupling in monkeys than in mice seems at first sight at odds with our conclusions that monkeys appear to be in a more strongly coupled regime than mice. It is possible that the physiological strengths of the connections are stronger in monkeys, which, when combined with stronger activation levels, can generate effective coupling strength equal to or larger than that of mice.

Running strongly enhances visual responses in the mouse V1.[50] Recent experiments found that, in the V1 of a species of monkey (the marmoset), running does not increase the mean response to visual stimuli but strongly modulates single-neuron responses, with a large fraction of cells suppressed.[51] This is similar to the difference we have found in mice vs. monkeys in responses to optogenetic stimuli (although the mouse mean response increase for running is considerably stronger), suggesting possible common mechanisms: the weaker running modulation in monkeys than in mice could, at least in part, be a consequence of stronger coupling in monkeys.

Relations with previous experimental works

The response properties of cortical networks to optogenetic inputs we have found here might also be relevant to understanding responses of neurons observed in other conditions. Large numbers of suppressed cells in response to optogenetic activation of pyramidal cells have been reported in experiments using one-photon stimulation in the mouse[52] and ferret[53] V1 or targeted two-photon stimulation of small ensembles in the mouse barrel cortex.[54] Weak changes in mean population activity, together with large changes in individual neuronal firing rates, also characterize the response of the cortex to some modulatory inputs,[55] e.g., in behavioral modulations of activity by head movements in mice[56] and running in marmosets.[51] Similar changes are seen after learning in experiments involving brain-machine interfaces.[57]

Recent works have reported significant increases in mean firing rate with optogenetic stimulation of E cells in the monkey V1.[58,59] Many factors might contribute to differences from our results, including strength, timing, and spatiotemporal structure of optogenetic stimulation, visual stimulus strengths, match of stimuli to neuronal preferences, recording methods, and behavioral paradigms. Andrei et al.[58] used very weak stimulus contrast, for which our data and models indicate weaker suppression, and used orientation-specific optogenetic stimulation, which, like tuned visual input, may drive mean activity. Chen et al.[59] used stronger optogenetic stimuli than in our experiments, as judged by the ratio of responses to purely optogenetic vs. purely visual stimulation. They found that when visual and optogenetic responses were of similar strength, visual response above optogenetic response was reduced compared with visual response alone, while when optogenetic responses were much stronger, visual response above optogenetic response was largely or completely eliminated. This seems similar to our results in that, in both experiments, the difference between the response to the combined stimulus and the response to the stronger stimulus is suppressed. Both the experiments in Andrei et al.[58] and in Chen et al.[59] involved behavioral tasks. In mice trained to detect optogenetic stimuli, network optogenetic responses are dramatically increased[60]; behavioral training might also alter network responses. Strong mean optogenetic responses were also recently reported in the monkey motor cortex.[8] Differences include cortical area, multiunit vs. single unit recording, and reduced light attenuation[61] (stronger optogenetic stimulation) via use of a fiber optic or light guide.

Limitations and future directions

For the best-fit match to the monkey at high contrast, the unstructured model had a ratio of optogenetic to E visual input of about 5×10^{-3}. The structured model relaxed this requirement, but its ratio for visual-stimulus-matched cells was still small, about 2×10^{-2}. Are these plausible? In layer 4 of the mouse V1, peak visually evoked E current was estimated to be 60–150 pA across cells.[62,63] In cultured neurons, optogenetically evoked currents are 100s to 1,000s of pA,[64] but light is severely attenuated with transmission through the cortex, by 50% over about 40 μm.[61] Thus, it is difficult to assay whether the model's requirement is met *in vivo* in cortical cells.

It will be interesting to examine the effects of other forms of structure, such as neurons with different preferred spatial phases[37,65] or preferred directions,[30] and to determine whether these might further relax the requirements on strength of coupling or optogenetic inputs.

Conclusions

The more than 70 million years of evolution separating mice and monkeys[66] has produced substantial differences in the structure of their cerebral and, in particular, visual cortices, e.g., in the presence of orientation and ocular dominance columns in primates,[33] but not in rodents,[34,67,68] the size[69] and structure[70] of their receptive fields and their connection statistics.[49] Our analysis of optogenetic modulation of visual response in the V1 shows that, despite these differences, mice and monkeys may share a common, remarkable resistance of their cortex to untuned optogenetic inputs at the population level, in spite of large changes in single-neuron firing rates. At the same time, we find a key difference may be that monkeys are in a more strongly coupled regime of higher firing rates, with correspondingly stronger resistance. These results might more generally yield insight into circuit mechanisms underlying differences in the effects of sensory or driving vs. modulatory inputs on cortical responses.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.neuron.2023.09.018.

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AUTHOR CONTRIBUTIONS

A.S., A.P., T.H.N., K.D.M., and N.B. designed the research and regularly discussed ongoing progress and plans. J.L., J.J.N., J.H.R., and M.H.H. provided the experimental data. A.S. conducted the analysis of the experimental data. A.S., A.P., and T.H.N. performed analytical calculations and numerical simulations of the unstructured model. A.P. and T.H.N. performed analytical calculations and numerical simulations of the structured model. A.S., A.P., T.H.N., K.D.M., and N.B. wrote the manuscript. All authors discussed the results and commented on the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

One or more of the authors of this paper self-identifies as an underrepresented ethnic minority in their field of research or within their geographical location. One or more of the authors of this paper self-identifies as a gender minority in their field of research. One or more of the authors of this paper self-identifies as a member of the LGBTQIA+ community. One or more of the authors of this paper received support from a program designed to increase minority representation in their field of research.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
Lentivirus (lenti-CaMKIIα-C1V1E162T-ts-EYFP) encoding the depolarizing opsin C1V1	Salk Institute viral vector core using constructs provided by Dr. Karl Deisseroth	N/A
Experimental models: Organisms/strains		
Emx1-cre mice (P40 to 1 year old)	The Jackson Laboratory	RRID:IMSR_JAX:005628
Two male rhesus macaque monkeys (wild type, 9 and 19 years old)	N/A	N/A
Deposited data		
Data analysis and network simulation codes	This paper	https://github.com/sanzeni/Reshuffling/tree/main
Software and algorithms		
Mworks	The Mworks Project	mworks.github.io
Scipy	Virtanen et al.[71]	https://docs.scipy.org/doc/scipy/index.html
Scikit-Learn	Pedregosa et al.[72]	https://scikit-learn.org/stable/index.html
Original code for modeling and analysis	This paper	https://doi.org/10.5281/zenodo.8329092
Other		
C and B Metabond	Parkell	S380
Kwik-sil	World Precision Instruments	KWIK-SIL

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Nicolas Brunel, nicolas.brunel@duke.edu

Materials availability

No new unique reagents or mouse lines were generated in this study.

Data and code availability

- Data analyzed in this paper are available from https://github.com/sanzeni/Reshuffling/tree/main.
- Data analysis and network simulation codes are available from https://github.com/sanzeni/Reshuffling/tree/main. All codes have been deposited at Zenodo, https://doi.org/10.5281/zenodo.8329092.
- Any additional information required to reanalyze the data reported in this work paper is available from the Lead Contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Mice

Data was collected from four Emx1-Cre adult (P40 to 1 year old) mice of both sexes (https://www.jax.org/strain/005628) at Harvard Medical School and all animal procedures were performed in accordance with the relevant animal care committee's regulations, and in accordance with NIH standards. More experimental details are available in Histed.[13]

Monkeys

A limitation of this study is that the neurophysiological data recorded from monkeys was obtained from only two rhesus macaques (Macaca mulatta), both adult (9 and 19 year old) and male. This is a consequence of the need to minimize the number of monkeys used, as reflected in the standard in the field for primate neurophysiological studies, which is to record from a sufficiently large number of neurons within each animal to obtain the statistical power needed to rigorously test hypotheses, and to replicate each finding in 2 monkeys. Given the small number of animals tested (N = 2) it is not feasible to detect sex differences, so it is unknown if the patterns observed here apply to female macaques. Experimental and surgical procedures were approved by the Salk Institute Institutional Animal Care and Use Committee and conformed to NIH guidelines for the care and use of laboratory animals.

METHOD DETAILS

Mice

A detailed description of the experimental methods is given in Histed.[13] In brief, neurophysiological data from Emx1-Cre animals ($n = 4$) were collected. Animals kept on a monitored water schedule were given small drops of water (~1 μl) every 60–120 s during recording to keep them awake and alert. The visual stimulus, a Gabor patch with spatial frequency 0.1 cycle/deg and sigma 12.5 deg, were presented for 115 ms [full width at half maximum (FWHM) intensity], and successive visual stimuli were presented every 1 s with contrasts varying between $[0 - 90]$%. Optogenetic light pulses were delivered on alternating sets of 10 stimulus presentations (light onset 500 ms before first stimulus; offset 500 ms after end of last stimulus; total light pulse duration 10.2 s). A 1-s delay was added after each set of 10 stimulus presentations. Extracellular probes were 32-site silicon electrodes (Neuronexus, probe model A4x8). Channelrhodpsin-2 (ChR2) was expressed in E neurons (as described in Histed and Maunsell[73]) using viral (AAV-EF1a-DIO-ChR2-mCherry, serotype 2 or 8; http://openoptogenetics.org) injections into the Emx1-Cre (Gorski et al.[74]; #5628, Jackson Laboratory) line. Virus (0.25–1.0 μl) was injected into a cortical site whose retinotopic location was identified by imaging autofluorescence responses to small visual stimuli. Light powers used for optogenetic stimulation were 500 μW/mm2 on the first recording session; in later sessions, dural thickening was visible and changes in firing rate were smaller, so power was increased (maximum 3 mW/mm2) to give mean spontaneous rate increases of approximately ~5 spk/s in that recording session. Optogenetic light spot diameter was 400–700 μm (FWHM) as measured by imaging the delivered light on the cortical surface.

Spike waveforms were sorted after the experiment using OfflineSorter (Plexon, Inc.). Single units (SU) were identified as waveform clusters that showed clear and stable separation from noise and other clusters, unimodal width distributions, and interspike interval histograms consistent with cortical neuron absolute and relative refractory periods. To compute neurons' visual responses, we counted spikes over a 175-ms period beginning 25 ms after stimulus onset. Zero-contrast responses were computed from the 175 ms period immediately preceding stimulus onset.

Monkeys

A detailed description of the experimental methods is given in Nassi et al.[14] In brief, two adult male rhesus macaques (Macaca mulatta) were each implanted with a custom titanium head post and silicone-based artificial dura recording chamber over V1. We injected a VSVg-pseudotyped lentivirus carrying the C1V1-EYFP gene behind the 1.3 kb CaMKIIα promoter (lenti-CaMKIIα-C1V1E162T-ts-EYFP; titer 3 × 1010 TU/ml) into a single location in V1 in each of the two monkeys (monkeys A and M) while they were anesthetized and secured in a stereotactic frame. Injections of the same viral construct were made into V1 of one additional monkey in order to assess specificity of viral expression to E neurons. In a second, distant location in V1 of monkey A, we injected an adeno-associated virus carrying the eArch3.0-EYFP gene behind the 1.3 kb CamKIIα promoter (AAV5-CamKIIα-eArch3.0-ts-EYFP; titer 4 × 1012 VP/ml).

Monkeys were alert and head restrained during all experiments. Single- and multi-unit waveforms were recorded using thin glass-coated tungsten electrodes (Alpha Omega, Engineering Inc., Nazareth, Israel). After isolating a single-unit or multi-unit cluster, we first assessed sensitivity to optogenetic stimulation. We randomly interleaved different stimulation intensities. Stimulation on each trial was continuous and lasted for 200 ms. Each condition was repeated at least five times. For a subset of light-sensitive units, we proceeded to measure responses to simultaneous optogenetic stimulation and visual stimulus presentation. The presented visual stimuli consisted of circular patches of drifting sinusoidal gratings whose size was matched to each recorded unit's preferred size (median diameter = 0.63deg), of mean luminance matching the surround (42 cd-m2) at the optimal orientation and spatial and temporal frequencies. Each stimulus condition was presented at least five times. For simultaneous optogenetic stimulation and visual stimulus presentation, we varied the contrast of the presented gratings in log steps (0%, 6%, 12%, 25%, 50%, and 99%) and stimulated with four different intensities including zero. Visual contrast and stimulation intensity were randomly interleaved. Optogenetic stimulation onset occurred simultaneously with the onset of the visual stimulus and lasted 200ms. The visual stimulus was always presented for at least 200ms, but in some cases lasted 300ms. Response was measured as the mean response over the interval 100–200ms after stimulus onset. Here, we show results only for two optogenetic intensities, zero and the largest intensity, but similar results hold also for intermediate intensities.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data analysis

The firing rates of individual neurons were computed averaging over trials. Error bars of example cells shown in Figures 1 and 7 represent the standard error of the mean across trials. The mean and standard error of quantities shown in Figures 2, 3, 5, and S6A–S6D were computed through a bootstrapping procedure. Specifically, we generated 10^3 ensemble of cells by random sampling with repetitions neurons measured experimentally, each ensemble consisted of the same number of neurons as those measured in the experiments. Within each ensemble, we measured various quantities of interest such as mean visual response, mean optogenetic response, and normalized covariance. By computing the mean and standard deviation over the 10^3 ensembles, we were able to estimate the overall mean and standard error of these quantities.

Mathematical methods

Network models

For all models, the single neuron response function (firing rate vs input current relationship) was taken to be the f-I curve of leaky integrate-and-fire neurons driven by white noise.[18–20,24] Specifically, the firing of a neuron belonging to population $A \in [E, I]$ in response to an input x, is given by

$$r_A = \phi_A(x) = \left[\tau_{rp} + \tau_A\sqrt{\pi}\int_{u_{min,A}}^{u_{max,A}} e^{u^2}(1+\mathrm{erf}(u))\,du\right]^{-1} \quad \text{(Equation 4)}$$

with

$$u_{max,A} = \frac{\theta - x}{\sigma_A}, u_{min,A} = \frac{V_r - x}{\sigma_A} \quad \text{(Equation 5)}$$

In Equations 4 and 5, τ_{rp} indicates the single neuron refractory period so that $1/\tau_{rp}$ is the maximal single neuron firing rate; τ_A is the membrane time constant; σ_A is a parameter controlling the smoothness of the transfer function. Throughout the paper, we assumed the values $\tau_{rp} = 2$ ms, $\theta = 20$ mV, $V_r = 10$ mV, $\tau_{E,I} = 20, 10$ ms and $\sigma_{E,I} = 10$ mV.

Indicating with r_A^i and μ_A^i the rate and the total input (recurrent + external) of the ith cell in population $A \in [E,I]$, activity in the network evolves in time according to the equation

$$\tau_A \frac{dr_A^i}{dt} = -r_A^i + \phi_A(\mu_A^i), \qquad \mu_A^i = \sum_{B \in [E,I,X]} \sum_{j=1}^{N_B} c_{AB}^{ij} W_{AB}^{ij} \tau_A r_B^j \quad \text{(Equation 6)}$$

Here c_{AB}^{ij} is the adjacency matrix (equal to 1 or 0 according to whether or not there is a connection from cell j of type B to cell i of type A) and N_B is the number of neurons in population B, where (X) indicates the externally driven input, *i.e.* visual input. We always use $N_X = N_E = 4N_I$.

Throughout the paper, we assumed uniform W's, $W_{AB}^{ij} = W_{AB}$, where the weights W_{AB} are parameterized as follows:

$$\begin{aligned} &W_{EE} = J && W_{EI} = Jg_E \\ &W_{IE} = J/\beta && W_{II} = Jg_I/\beta \\ &W_{EX} = (G_I g_E N_I/N_E - G_E)W_{EE} && W_{IX} = (G_I g_I N_I/N_E - G_E)W_{IE} \end{aligned} \quad \text{(Equation 7)}$$

These definitions were chosen so that the parameters G_I and G_E give the gain of the E and I population in the strong coupling limit (see Sanzeni et al.[24]).

With optogenetic input, the network dynamics become

$$\begin{aligned} \tau_A \frac{d(r_A^i + \Delta r_A^i)}{dt} &= -r_A^i - \Delta r_A^i + \phi_A(\mu_A^i + \Delta\mu_A^i + \lambda_A^i) \\ \Delta\mu_A^i &= \sum_{B \in [E,I,X]} \sum_{j=1}^{N_B} c_{AB}^{ij} W_{AB}^{ij} \tau_A \Delta r_B^j \end{aligned} \quad \text{(Equation 8)}$$

where λ_A^i represents the optogenetic input of the ith neuron in the network, which varies according to the opsin expression in that neuron, while Δr_A^i and $\Delta\mu_A^i$ indicate the change in rate and in the recurrent inputs produced by optogenetic stimuli.

Unstructured models

We investigated large randomly connected networks of E and I rate neurons. In the theoretical analysis of the model (mean field theory described below and simulations shown in Figures 4 and S2), the network connectivity was assumed to follow an Erdös–Rényi statistics, *i.e.* each element of the adjacency matrix c_{AB}^{ij} was 1 with probability $p = 0.1$ and 0 otherwise. The mean and variance of the number of pre-synaptic neurons from population B per cell are given by $K_B = pN_B$ and $\sigma_{K_B}^2 = (1-p)K_B$, respectively. We define $K \equiv K_E = pN_E$, and use the notation $\gamma = K_I/K_E$. In fitting the model to data (Figure 5), we considered more general statistics by generating c_{AB}^{ij} as follows: for each row i, a random number, K_i (Gaussian distributed, with mean K_B and standard deviation $\sigma_{K_B} = CV_K \times K_B$) of elements were set to 1, with autapses not allowed, and the other elements were set to 0 (where CV_K is a constant that is identical for $B = E$, I, or X).

Structured models

For the structured models, we place cells onto evenly spaced discrete locations on a 25×25 periodic grid, so that there are $N_G = 625$ grid points, with 8 E and 2 I cells per grid location. The grid is a square of side $dl = 1$ mm. For the monkey, we introduce a smooth aperiodic orientation map using a method similar to that of Kaschube et al.[36] by summing 500 plane waves of the form

$\exp[i(k\cos(j\pi/500)x + k\sin(j\pi/500)y + \phi_j)]$ for $j = 0, 1, \ldots, 499$. The wavenumber k is chosen to be $6\pi/dl$ such that the wavelength is $dl/3$, and the phase ϕ_j is chosen as a random angle for each j. The preferred orientation at a point in the grid is calculated as half the phase of the sum of plane waves at that point, such that the orientations are between 0° and 180°. For the mouse, we generate a salt-and-pepper orientation map by randomly shuffling a smooth orientation map generated as described above. We let $\theta(\mathbf{x})$ be the preferred orientation at two-dimensional grid position $\mathbf{x}$.

The probability of a connection from a cell at position $\mathbf{y}$ of type A, $A \in E, I$, to any cell at position $\mathbf{x}$ is computed as follows. We let $d(\mathbf{x}, \mathbf{y})$ be the shortest distance across the periodic grid between $\mathbf{x}$ and $\mathbf{y}$, and $d\theta(\theta(\mathbf{x}), \theta(\mathbf{y}))$ be the shortest distance around the 180° circle of orientations between $\theta(\mathbf{x})$ and $\theta(\mathbf{y})$. We let n_A be the number of cells of type A at a single grid position ($n_E = 8$, $n_I = 2$). We then form the functions $\tilde{g}_l(\mathbf{x}, \mathbf{y}, A) = e^{-d(\mathbf{x},\mathbf{y})^2/(2S_{lA}^2)}$ and $\tilde{g}_{ori}(\mathbf{x}, \mathbf{y}, A) = e^{-d\theta(\theta(\mathbf{x}),\theta(\mathbf{y}))^2/(2S_{ori,A}^2)}$, and normalize each by its mean to obtain the product

$$g(\mathbf{x}, \mathbf{y}, A) = \left(\frac{\tilde{g}_l(\mathbf{x}, \mathbf{y}, A)}{\sum_{x'y'} \tilde{g}_l(\mathbf{x}', \mathbf{y}', A) \Big/ N_G^2}\right)\left(\frac{\tilde{g}_{ori}(\mathbf{x}, \mathbf{y}, A)}{\sum_{x'y'} \tilde{g}_{ori}(\mathbf{x}', \mathbf{y}', A) \Big/ N_G^2}\right)$$

We then choose the number of connections $K_A^i(\mathbf{x})$ for the i^{th} cell at $\mathbf{x}$ from a Gaussian distribution with mean $K_A(\mathbf{x}) = pn_A \sum_{\mathbf{y}} g(\mathbf{x}, \mathbf{y}, A)$ and standard deviation $CV_K \times K_A(\mathbf{x})$, where p is a constant that is chosen such that the mean number of excitatory connections is similar to that of the unstructured model (for excitatory connections, ≈525 in our mouse models, ≈ 508 in our monkey models). We choose a total of $K_A^i(\mathbf{x})$ presynaptic neurons (excluding autapses) by a weighted random sample without replacement, with weights $g(\mathbf{x}, \mathbf{y}, A)/\left(\sum_{y'} n_A g(\mathbf{x}, \mathbf{y}, A)\right)$ (choice function from python package numpy.random; if there is a potential autapse, it is removed from the sum in the denominator). The visual input to a cell of type A at $\mathbf{x}$ from a stimulus centered at $\mathbf{y}$ with orientation θ_{stim} is $W_{AX}K_X r_X e^{-d(\mathbf{x},\mathbf{y})^2/(2S_{stim}^2)} e^{-d\theta(\theta(\mathbf{x}),\theta_{stim})^2/(2S_{tune}^2)}$. Here, $K_X = 0.1N_X$ is the number of visual inputs to a cell and r_X their firing rate. We set the input to have zero variance for simplicity. When we refer to visual-stimulus-matched cells, we mean cells with orientation tuning within 22.5° and receptive field within $0.6dl$ of the visual stimulus.

Network simulations

Simulations of the network dynamics were performed in python. For each parameter set, we generated one realization of the network structure (adjacency matrices c_{AB}^{ij}, firing of the external population r_X^i (generated from a Gaussian distribution with mean r_X and standard deviation $\sigma_X = 0.2\, r_X$), opsin expression λ_A^i (generated from a lognormal distribution with mean λ and standard deviation σ_λ) and simulated the network dynamics given by Equations 6 and 8 using the explicit Runge-Kutta method of order 5 (implemented with the function solve_ivp of the python package scipy.integrate). The network dynamics was run for a total simulated time of $200\tau_E = 4$ s for the unstructured model and $300\tau_E = 6$ s for the structured model and computed rates were stored every $\tau_I/3 = 3.33$ ms. The firing rate of a cell was obtained by measuring its average rate over the simulation time (excluding the initial $10\tau_E = 0.2$ s for the unstructured model and $100\tau_E = 2$ s for the structured one). To measure optogenetic responses, we simulated the same realization of network structure twice, once with $\lambda_A^i \neq 0$ and once with $\lambda_A^i = 0$. When calculating optogenetic response statistics, we only include cells whose response was higher than 1 spk/s in either the simulation with $\lambda_A^i = 0$ or the simulation with $\lambda_A^i \neq 0$. Parameters for all simulations are given in Table of Parameters of Unstructured Models and Table of Parameters of Structured Models, and resulting weight strengths in Table of best-fit-model weights.

Fitting procedure

For both unstructured and structured models, we fit both the network parameters and the mapping between contrasts and external input strengths to the data by numerically simulating model networks with randomly sampled parameters and external input strengths. We then train a multilayer perception (MLPRegressor from python package sklearn.neural_network[72]) to learn a mapping $f_Y(r_X, \Theta)$ between network parameters Θ and an input strength r_X to population response statistics $Y \in \{r, \Delta r, \sigma_r, \sigma_{\Delta r}, \mathrm{Cov}(r, \Delta r)\}$. We can use this mapping to also calculate $f_Y(r_X, \Theta)$ for other statistics such as $Y \in \{r^L = r + \Delta r, \sigma_{r^L} = \sqrt{\sigma_r^2 + \sigma_{\Delta r}^2 + 2\mathrm{Cov}(r, \Delta r)}, \rho = \mathrm{Cov}(r, \Delta r)/\sigma_{\Delta r}^2\}$

From our data analysis, we calculate the mean $\widehat{Y}_c$ and standard error $\sigma_{\widehat{Y}_c}$ of the quantities $Y \in \{r, r^L, \sigma_r, \sigma_{r^L}, \sigma_{\Delta r}, \rho\}$ from the firing rates of recorded neurons at contrast c. We define the contrast-specific loss $\mathcal{L}_c(r_X, \Theta)$ as the mean square error weighted by the standard errors between the population response statistics $\widehat{Y}_c$ at a given contrast c and the predicted statistics from the model given by the multilayer perceptron mapping $f_Y(r_X, \Theta)$. For a given set of network parameters Θ, we also define the total loss function across all contrasts $\mathcal{L}(\Theta)$ as the minimum of $\sum \mathcal{L}_c(r_{X,c}, \Theta)$ as a function of strictly increasing mappings between contrast and input strength $\{r_{X,c} | r_{X,c-1} < r_{X,c}\}$ (least_squares function from python package scipy.optimize, using the default trust region reflective method). The loss functions can be summarized as

$$\mathcal{L}_c(r_X, \Theta) = \sum_{Y \in \{r, r^L, \sigma_r, \sigma_{r_L}, \sigma_{\Delta r}, \rho\}} \frac{1}{2}\left(\frac{\widehat{Y}_c - f_Y(r_{X,c}, \Theta)}{\sigma_{\widehat{Y}_c}}\right)^2 \qquad \text{(Equation 9)}$$

$$\mathcal{L}(\Theta) = \min_{\{r_{X,c}\}} \sum_c \mathcal{L}_c(r_{X,c}, \Theta) \qquad \text{(Equation 10)}$$

Finally, we perform least squares regression (again, using the least_squares function from scipy.optimize using the default method) on $\mathcal{L}(\Theta)$ to find candidate best-fit model parameters Θ^*.

We do not fit directly to the mean-field results for three reasons. Firstly, calculating the covariance between the baseline rates and the changes in rates requires performing a numerically costly triple-integral over the log-normal distributed opsin expression λ_A^i and the joint-Gaussian distributed baseline input μ_A^i and change in recurrent input $\Delta\mu_A^i$. For fixed parameters of the optogenetic input or over a small interval of the parameters we can pre-compute quantities $\int d\lambda_A^i P(\lambda_A^i)\phi_A[\lambda_A^i + \mu_A]$ and $\int d\lambda_A^i dx P(\lambda_A^i) P_{\mathcal{N}}(x)\phi_A[\lambda_A^i + \mu_A + x\sigma_{\mu_A}]$ over many values of μ_A and σ_{μ_A} and interpolate between those pre-computed values to speed up the mean-field calculations. However, accurately fitting the mean-field results over the entire range of considered optogenetic input parameters is expensive compared to training the perceptron. Secondly, due to finite-size effects from our choice of a mean number of excitatory connections $K = 500$ there can be a discrepancy between the mean-field result compared to the simulation results (see Figure S2). Lastly, there does not exist an accurate mean-field approximation for the structured networks that would allow us to fit them to the data. In order to have similar fitting procedures between the two models, we choose to use the perceptron-based fitting method for both.

Unstructured model

Predictions of the model depend on a list of unknown parameters (indicated in what follows as Θ) related to: connectivity ($J, K, \beta, g_E, g_I, CV_K$); external inputs ($G_E, G_I$ and one value of r_X per each contrast); and optogenetic inputs (λ, σ_λ). To fit these parameters to experimental data, we assumed $G_I = 2G_E$, consistent with the fact that in vivo, I firing rates are typically about twice E firing rates (e.g., Sanzeni et al.[4]). This choice is motivated by the fact that experimental recordings do not distinguish cell types; this limitation makes our model under-constrained by the data (since the same average firing could be obtained by an infinite combination of firing rates of E and I cells). Without loss of generality, we assumed $G_E = 1$ (any other choice can be obtained by rescaling r_X) and $K = 500$ (any other choice can be obtained by rescaling $J \sim 1/\sqrt{K}$). Moreover, we took $\sigma_X = 0$, and controlled variability in the external input by varying CV_K.

With the above assumptions, model parameters were inferred as follows. First, we measured optogenetic responses in a large number (770195) of network simulations; each one obtained assuming model parameters generated randomly and uniformly in the intervals

$$\begin{array}{ll} g_E \in [3,7] & g_I \in [2, g_E - 0.5] \\ \log_{10}(\beta) \in [-1,1] & \log_{10}(\sigma_\lambda/\lambda) \in [-1,1] \\ \log_{10}(J) \in [-5,-3] & \log_{10}(r_X) \in [-0.3, 1.7] \\ \log_{10}(\lambda) \in [-0.3, 1.7] & \log_{10}(CV_K) \in [-3.5, -0.5] \end{array} \qquad \text{(Equation 11)}$$

Parameter regions leading to strongly chaotic dynamics (auto-correlation time of neural activity shorter than one second) were excluded from the fitting procedure for the unstructured model. Specifically, all moments of the network response for which the simulated autocorrelation time was smaller than 1 second were set to 10^{10}, preventing the least square optimization algorithm from exploring regions of the parameter space that lead to strong chaotic dynamics. Chaotic dynamics leads to large fluctuation of responses in network models from one realization of the network connectivity to the other. This property significantly complicates predictive modeling, as numerous simulations are required for the multilayer perceptron to accurately predict network responses across various parameters. We note that excluding strong chaos does not prevent the best-fit networks from being weakly chaotic, as seen in Figure 5.

Starting from the results of these simulations, we trained a multilayer perceptron (MLPRegressor from python package sklearn.neural_network[72]; specifics were activation=ReLU, hidden_layer_sizes=(100, 150, 50), with all other hyperparameters at their default values) to predict the mapping $f_Y(r_X, \Theta)$ between model parameters (Θ) and a feedforward input strength (r_X) to moments of the optogenetic response The learned multilayer perceptron was able to predict the outcome of simulations with high accuracy (with 20% of held-out simulations, the crossvalidated performance had $R^2 = 0.924$). We then fitted model parameters Θ to data by minimizing $\mathcal{L}(\Theta)$ defined in Equation 10 starting from 100 random initial parameters Θ_0 generated within the same intervals as above, restricting the least-squares regression to remain in these intervals. We then choose the one model from the 100 candidate best-fit model parameters Θ^* with the smallest predicted loss value.

Structured model

In order to fit the structured model, we allowed G_I to vary since the spatial structure often causes the network to be more inhibition dominated than the unstructured model. As for the unstructured model, we measured optogenetic responses in a large number (484848 with salt-and-pepper maps and 508719 with smooth maps) of network simulations; we generated parameters randomly and uniformly in the following intervals

$$
\begin{array}{ll}
G_I \in [1,2] & \log_{10}(\sigma_\lambda/\lambda) \in [-0.5, 1.5] \\
S_{I,E} \in [0,1] & S_{I,I} \in [(2/3)S_{I,E}, 2S_{I,E}] \\
S_{ori,I} \in [20,60] & S_{tune} \in [10,40]
\end{array}
\qquad \text{(Equation 12)}
$$

In addition, we excluded parameters where the parameterization from Equation 7 would give negative mean external input to any population. We fixed $S_{ori,E} = 30^\circ$ and $S_{stim} = 0.5dl$. All other parameters not listed above are generated with the same intervals as for the unstructured model. For each simulation, we also generate a different random orientation map.

Due to the relatively coarse grid size (25x25 microcolumns), there can be some variability between simulation results for fixed model parameters but with different orientation maps. Ideally, we would average results over many orientation maps to have more accurate moments of the optogenetic response for training the multilayer perceptron, but doing so over all of our simulated models is expensive. In order to reduce the required computational cost, we only average over many orientation maps (choosing four fixed maps to average over) for sampled model parameters Θ and input strengths r_X where $\mathcal{L}_c(r_X,\Theta) < 3$ for any contrast c. Next, we explore the parameter space near these models whose optogenetic responses are already similar to the data by perturbing their parameters and repeating the map-averaging process. For each of the previous map-averaged models, we repeatedly generate a new perturbed model by multiplying each parameter by a random value in the interval $[0.85, 1.15]$ and simulating their responses over the same four orientation maps. None of our map-averaged models had strong chaotic dynamics, and we did not need to exclude additional regions of parameter space from the fitting procedure.

Finally, we train the multilayer perceptron (MLPRegressor,[72] described above; specifics were activation=ReLU, hidden_layer_sizes=(150, 100, 50), alpha= 10^{-5}, learning_rate_init=0.002, tol= 10^{-6}, beta_2=0.99, with all other hyperparameters at their default values) on all orientation-map-averaged simulation results to again predict the mapping $f_Y(r_X,\Theta)$ between model parameters and moments of the optogenetic response. We use this learned mapping to again fit model parameters Θ by minimizing $\mathcal{L}(\Theta)$. Since we only trained the multilayer perceptron with model parameters that represent a subset of the parameter intervals, we choose our 100 initial parameters Θ_0 out of the model parameters that were used to train the multilayer perceptron. For each of the 100 candidate best-fit model parameters Θ^*, we simulate the network at the second largest contrast c^* and choose the best fit model as the one with the smallest value of $\mathcal{L}_{c^*}(r_{X,c^*},\Theta*)$.

Table of Parameters of Unstructured Models

			Figure 4	Figure 5
Parameter	Symbol	Units	(weak, strong)	(comb., mice, monk.)
Mean E connections per neuron	K		2000	500
CV of in-degree	CV_K		(1-p)/$\sqrt{K}$	(0.015, 0.020, 0.007)
Connection probability	p		0.1	0.1
Number of excitatory neurons	N_E		K/p	K/p
Ratio of I to E Population Size	γ		0.25	0.25
E time constant	τ_E	ms	20	20
I time constant	τ_I	ms	10	10
Synaptic efficacy	J	mV	(0.01, 0.3)	(0.14, 0.13, 0.44)
Ratio of I-to-E over E-to-E	g_E		8	(4.2, 9.97, 4.58)
Ratio of I-to-I over E-to-I	g_I		3	(3.24, 8.4, 2.73)
Ratio of E-to-E over E-to-I	β		1	(0.17, 0.33, 0.81)
Gain of the E population	G_E		from Equation 7	1
Gain of the I population	G_I		from Equation 7	2
Strength of FF input to E	W_{EX}		$3J$	from Equation 7
Strength of FF input to I	W_{IX}		($0.5J$,$2.5J$)	from Equation 7
Mean FF input rates	r_X	spk/s	20	([13.5 - 23.4] &[30.9 - 77.6], [7.24 - 13.8], [14.45 - 47.86])
Std FF input rates	σ_X	spk/s	$0.2r_X$	0
Mean opto. input strength	λ	mV	20	(1.9 & 5.5, 4.2, 2.5)
CV of opto. strength	σ_λ/λ		1	(1.69, 0.79, 6.6)

Parameters used in Figures 4 and 5. In tables of parameters, "FF input" means feedforward input, i.e., external (visual) input.

Table of Parameters of Structured Models

Parameter	Symbol	Units	Mouse	Monkey
Mean FF E connections per neuron	K		500	500
CV of in-degree	CV_K		0.0034	0.0040
Mean recurrent connection probability	p		0.09	0.09
Number of grid locations	N_l		25 × 25	25 × 25
Number of E neurons per grid location	N_E		8	8
Ratio of I to E Population Size	γ		0.25	0.25
E time constant	τ_E	ms	20	20
I time constant	τ_I	ms	10	10
Synaptic efficacy	J	mV	0.093	0.526
Ratio of I-to-E over E-to-E	g_E		8.09	3.89
Ratio of I-to-I over E-to-I	g_I		2.58	2.40
Ratio of E-to-E over E-to-I	β		0.34	1.60
Gain of the E population	G_E		1	1
Gain of the I population	G_I		1.80	1.68
Strength of FF input to E	W_{EX}		from Equation 7	from Equation 7
Strength of FF input to I	W_{IX}		from Equation 7	from Equation 7
Mean FF input rates	r_X	spk/s	[10.07 - 23.06]	[4.47 - 20.13]
Std FF input rates	σ_X	spk/s	0	0
Mean opto. input strength	λ	mV	5.53	9.01
CV of opto. strength	σ_λ/λ		0.71	2.53
Spatial grid length	dl	mm	1	1
Spatial width of FF input	S_{stim}	mm	0.5	0.5
Tuning width of FF input	S_{tune}	deg $^\circ$	48.36	20.83
Spatial width of recurrent E	$S_{l,E}$	Mm	0.256	0.948
Spatial width of recurrent I	$S_{l,I}$	Mm	0.136	0.840
Tuning width of recurrent E	$S_{ori,E}$	deg $^\circ$	30	30
Tuning width of recurrent I	$S_{ori,I}$	deg $^\circ$	21.77	31.71

Parameters used in Figure 5.

Table of best-fit-model weights

Model	$\begin{pmatrix} W_{EE} & -W_{EI} \\ W_{IE} & -W_{II} \end{pmatrix}$	$\begin{pmatrix} W_{EX} \\ W_{IX} \end{pmatrix}$
Unstructured Combined	$\begin{pmatrix} 0.14 & -0.59 \\ 0.82 & -2.67 \end{pmatrix}$	$\begin{pmatrix} 0.15 \\ 0.51 \end{pmatrix}$
Unstructured Mouse	$\begin{pmatrix} 0.13 & -1.30 \\ 0.39 & -3.31 \end{pmatrix}$	$\begin{pmatrix} 0.52 \\ 1.26 \end{pmatrix}$
Unstructured Monkey	$\begin{pmatrix} 0.44 & -2.02 \\ 0.54 & -1.48 \end{pmatrix}$	$\begin{pmatrix} 0.57 \\ 0.20 \end{pmatrix}$
Structured Mouse	$\begin{pmatrix} 0.093 & -0.75 \\ 0.27 & -0.71 \end{pmatrix}$	$\begin{pmatrix} 0.25 \\ 0.044 \end{pmatrix}$
Structured Monkey	$\begin{pmatrix} 0.53 & -2.05 \\ 0.33 & -0.79 \end{pmatrix}$	$\begin{pmatrix} 0.33 \\ 0.0026 \end{pmatrix}$

Weights used in best fit models for Figure 5, computed from parameters in previous tables and Equation 7.

Mean field theory for the unstructured model

For $N_B \gg 1$ in the unstructured model, μ_A^i and $\Delta\mu_A^i$ in Equations 6 and 8 are well approximated by correlated random Gaussian variables with

$$\mu_A = \tau_A \sum_{B\in[E,I,X]} W_{AB}K_B r_B$$
$$\sigma^2_{\mu_A} = \tau_A^2 \sum_{B\in[E,I,X]} W_{AB}^2 K_B\left[\sigma^2_{r_B} + (1-p)r_B^2\right]$$
$$\Delta\mu_A = \tau_A \sum_{B\in[E,I]} W_{AB}K_B \Delta r_B \quad \text{(Equation 13)}$$
$$\sigma^2_{\Delta\mu_A} = \tau_A^2 \sum_{B\in[E,I]} W_{AB}^2 K_B\left[\sigma^2_{\Delta r_B} + (1-p)\Delta r_B^2\right]$$
$$\mathrm{Cov}(\mu_A,\Delta\mu_A) = \tau_A^2 \sum_{B\in[E,I]} W_{AB}^2 K_B[\mathrm{Cov}(r_B,\Delta r_B) + (1-p)r_B\Delta r_B]$$

In Equation 13, μ_A ($\Delta\mu_A$) and σ_{μ_A} ($\sigma_{\Delta\mu_A}$) are the mean and variance of μ_A^i ($\Delta\mu_A^i$). The covariance $\mathrm{Cov}(\mu_A,\Delta\mu_A)$ is computed starting from.

$$\mu_A^i + \Delta\mu_A^i = \sum_{B\in[E,I,X]} \tau_A W_{AB} \sum_{j=1}^{N_B} c_{AB}^{ij}\left(r_B^j + \Delta r_B^j\right) \quad \text{(Equation 14)}$$

applying the identity $\mathrm{Var}\,(X+Y) = \mathrm{Var}\,(X) + \mathrm{Var}\,(Y) + 2\mathrm{Cov}(X,Y)$ to both sides and using the central limit theorem. Note that optogenetic input affects firing differently than external and recurrent inputs. In fact, firing depends on the whole distribution of λ_A^i but only on the first two moments of external and recurrent inputs.

If the network dynamics settle in a stable fixed point, the statistics of currents and rates in the network are related by

$$\begin{aligned} r_A^i &= \phi_A\left[\mu_A^i\right] \\ r_A^i + \Delta r_A^i &= \phi_A\left[\lambda_A^i + \mu_A^i + \Delta\mu_A^i\right] \end{aligned} \quad (15)$$

and moments of the rates can be computed by self-consistently solving a set of ten equations

$$\left\{\begin{aligned} r_A &= \int_{-\infty}^{\infty} d\mu_A^i P\left(\mu_A^i\right)\phi\left(\mu_A^i\right) \\ \sigma^2_{r_A} &= \int_{-\infty}^{\infty} d\mu_A^i P\left(\mu_A^i\right)\phi\left(\mu_A^i\right)^2 - r_A^2 \\ \Delta r_A &= \int_{-\infty}^{\infty} d\mu_A^i d\Delta\mu_A^i d\lambda_A^i P\left(\mu_A^i,\Delta\mu_A^i\right)P\left(\lambda_A^i\right)\left[\Delta\phi_A^i\right] \\ \sigma^2_{\Delta r_A} &= \int_{-\infty}^{\infty} d\mu_A^i d\Delta\mu_A^i d\lambda_A^i P\left(\mu_A^i,\Delta\mu_A^i\right)P\left(\lambda_A^i\right)\left[\Delta\phi_A^i\right]^2 - \Delta r_A^2 \\ \mathrm{Cov}(r_A,\Delta r_A) &= \int_{-\infty}^{\infty} d\mu_A^i d\Delta\mu_A^i d\lambda_A^i P\left(\mu_A^i,\Delta\mu_A^i\right)P\left(\lambda_A^i\right)\left[\phi\left(\mu_A^i\right) - r_A\right]\left[\Delta\phi_A^i - \Delta r_A\right] \end{aligned}\right. \quad \text{(Equation 16)}$$

with $\Delta\phi_A^i = \phi(\mu_A^i + \Delta\mu_A^i + \lambda_A^i) - \phi(\mu_A^i)$.

If the fixed point of the network dynamics is unstable, rates and currents become time-dependent. One can again use a Gaussian ansatz, in which the total inputs at different times ($\mu_A^i(s)$ and $\mu_A^i(t)$) are jointly Gaussian with means $\mu_A(s), \mu_A(t)$ and covariance $\sigma^2_{\mu_A}(s,t)$. Analogously, rates at different times ($r_A^i(s)$ and $r_A^i(t)$) are jointly Gaussian with means $r_A(s), r_A(t)$ and covariance $\sigma^2_{r_A}(s,t)$. At any given pair of time points t and s, the moments of rates and currents are related by Equation 13. The temporal evolution of the moments can be computed using standard techniques.[15] In particular, indicating with $C_{r_A}(s,t) = N_A^{-1}\sum_{i=1}^{N_A} r_A^i(s) r_A^i(t)$ the autocorrelation function of the rates of population A at times s and t, we can define the autocovariance function of the rates as

$$\sigma^2_{r_A}(s,t) = C_{r_A}(s,t) - r_A(s)r_A(t) \quad \text{(Equation 17)}$$

By multiplying Equation 8 at times s and t, averaging over cells in population A, and rearranging terms we see that $C_{r_A}(s,t)$ evolves in time according to

$$\left(\tau_A\frac{d}{ds} + 1\right)\left(\tau_A\frac{d}{dt} + 1\right)C_{r_A}(s,t) = C_{\phi_A[\mu_A]}(s,t)$$

We can repeat the above steps to derive equations for the evolution in time of all five order parameters per population of neurons (mean, autocovariance, and cross-covariance of rates without and with optogenetic stimuli). Defining

$$M_{\phi_A[\mu_A]}(t) = N_A^{-1}\sum_{i=1}^{N_A}\phi_A\left[\mu_A^i(t)\right]$$

and

$$M_{\phi_A[\tilde{\mu}_A]}(t) = N_A^{-1}\sum_{i=1}^{N_A}\phi_A\left[\mu_A^i(t) + \Delta\mu_A^i(t) + \lambda_A^i\right]$$

the order parameters evolve in time according to

$$\begin{aligned}
\left(\tau_A\frac{d}{dt}+1\right)r_A &= M_{\phi_A[\mu_A]}(t)\\
\left(\tau_A\frac{d}{dt}+1\right)\Delta r_A &= M_{\phi_A\left[\tilde{\mu}_A\right]}(t) - M_{\phi_A[\mu_A]}(t)\\
\left(\tau_A\frac{d}{ds}+1\right)\left(\tau_A\frac{d}{dt}+1\right)C_{r_A}(s,t) &= C_{\phi_A[\mu_A]}(s,t)\\
\left(\tau_A\frac{d}{ds}+1\right)\left(\tau_A\frac{d}{dt}+1\right)C_{\Delta r_A}(s,t) &= C_{\phi_A\left[\tilde{\mu}_A\right]}(s,t) - \mathrm{Corr}\left(\phi_A[\mu_A],\phi_A\left[\tilde{\mu}_A\right]\right)(s,t)\\
&\quad - \mathrm{Corr}\left(\phi_A\left[\tilde{\mu}_A\right],\phi_A[\mu_A]\right)(s,t) + C_{\phi_A[\mu_A]}(s,t)\\
\left(\tau_A\frac{d}{ds}+1\right)\left(\tau_A\frac{d}{dt}+1\right)\mathrm{Corr}(r_A,\Delta r_A)(s,t) &= \mathrm{Corr}\left(\phi_A[\mu_A],\phi_A\left[\tilde{\mu}_A\right]\right)(s,t) - C_{\phi_A[\mu_A]}(s,t)
\end{aligned} \qquad \text{(Equation 18)}$$

Numerical solution of mean field equations

We solve for the stationary statistics of the chaotic network by explicitly integrating Equation 18 in time until the order parameters converge to a steady state. We start with initial values for $r_A(0)$ and $\Delta r_A(0)$ as well as initial functions for $C_{r_A}(s,0) = C_{r_A}(0,t)$, $C_{\Delta r_A}(s,0) = C_{r_A}(0,t)$, and $\mathrm{Corr}(r_A,\Delta r_A)(s,0) = \mathrm{Corr}(r_A,\Delta r_A)(0,t)$. The means $r_A(t)$ and $\Delta r_A(t)$ are integrated with the usual explicit RK4 method, but the autocovariance and cross-covariance functions are integrated on a grid with an explicit Euler-like method derived using forward finite differences. For example, when integrating $C_{r_A}(s,t)$ we can express $C_{r_A}(s+dt,t+dt)$ in terms of $r_A(s)$, $r_A(t)$, $C_{r_A}(s,s)$, $C_{r_A}(t,t)$, $C_{r_A}(s,t)$, $C_{r_A}(s+dt,t)$, and $C_{r_A}(s,t+dt)$. Assuming that we have solved for the order parameters at all time points $(s \le t_0, t \le t_0)$, we can explicitly solve for the order parameters at time points $(s \le t_0+dt, t_0+dt)$ and $(t_0+dt, t \le t_0+dt)$. Furthermore, we can ignore portions of the grid for $|s-t|>T$ for some T that is much longer than the width of the steady state autocorrelation function in order to reduce the computation time required for the integration.

A useful simplification of the equations governing the evolution of the order parameters is to calculate quantities such as $C_{\phi_A[\mu_A]}(s,t)$ by setting the mean and variance of μ_A^i at times s and t equal to the mean and variance at time max(s,t). While this change increases the required integration time for the order parameters to reach the steady state, it reduces the number of dependent variables needed to calculate quantities like $C_{\phi_A[\mu_A]}(s,t)$ from five to three. With only three dependent variables, it becomes feasible to pre-calculate these autocorrelation functions on a 3D grid and interpolate between the pre-computed points in order to accelerate the integration. In addition, we can precompute quantities such as $M_{\phi_A[\mu_A]}(t)$, which only depends on the mean and variance of the inputs, on a 2D grid and interpolate them as well.

The cross-correlation function $\mathrm{Corr}(\phi_A[\mu_A],\phi_A[\tilde{\mu}_A])(s,t)$ cannot be pre-computed and interpolated since the means and variances of the cross-correlated variables are not equal. However, since we have already interpolated $M_{\phi_A[\mu_A]}(t)$ and $M_{\phi_A[\tilde{\mu}_A]}(t)$ we can efficiently calculate the cross-correlation function from a single Gaussian integral.

Indicating with $M_{\phi_A[\mu_A]}[\mu_A(t),\sigma^2_{\mu_A}(t,t)]$ and $M_{\phi_A[\tilde{\mu}_A]}[\mu_A(t)+\Delta\mu_A(t),\sigma^2_{\mu_A+\Delta\mu_A}(t,t)]$ the precomputed mean rates as a function of the means and variances of the inputs, we can express the cross-correlation function as

$$\begin{aligned}
\mathrm{Corr}\left(\phi_A[\mu_A],\phi_A\left[\tilde{\mu}_A\right]\right)(s,t) = \int_{-\infty}^{\infty} Dx M_{\phi_A[\mu_A]}\left[\mu_A(s)+\mathrm{sign}(\eta)x\sqrt{|\eta|}\ \sigma_{\mu_A}(s,s),(1-|\eta|)\sigma^2_{\mu_A}(s,s)\right]\\
\times M_{\phi_A\left[\tilde{\mu}_A\right]}\left[\mu_A(t)+\Delta\mu_A(t)+x\sqrt{|\eta|}\,\sigma_{\mu_A+\Delta\mu_A}(t,t),(1-|\eta|)\sigma^2_{\mu_A+\Delta\mu_A}(t,t)\right]
\end{aligned} \qquad \text{(Equation 19)}$$

with the Pearson correlation coefficient η defined as

$$\eta = \frac{\mathrm{Cov}(\mu_A,\mu_A+\Delta\mu_A)(s,t)}{\sigma_{\mu_A}(s,s)\sigma_{\mu_A+\Delta\mu_A}(t,t)}.$$